GAUSTF: An Object-Oriented Framework for Spatiotemporal Fusion of Satellite Data

Xudong Guan, Chong Huang, Wei Zhao, *Senior Member, IEEE*, Bo Kong, and Rui Zhang

***Abstract*—High spatiotemporal resolution remote sensing data is critical for monitoring land surface dynamics. While recent advancements, particularly in deep learning, have significantly pushed the boundaries of prediction accuracy, the field currently lacks highly abstracted mathematical models for fusion. Existing traditional model-driven methods often lack unified mathematical formulations, relying instead on rigid, pixel-wise procedures such as empirical window searching and filtering. This structural fragmentation inherently limits their theoretical interpretability. To bridge this gap, this study proposes a novel Gaussian model-based spatiotemporal fusion method (GAUSTF). GAUSTF unifies abstract statistical modeling with linear spectral unmixing by hypothesizing that temporal reflectance changes (RCs) of land-cover objects can be approximated by Gaussian distributions. Instead of performing complex pixel-to-pixel mapping, we treat mixed pixels as compositions of multiple normal distributions, transforming the fusion task into a parameter inference problem for object-level reflectance recovery. We evaluated GAUSTF on four benchmark datasets against classical algorithms like STARFM and FSDAF. The results indicate that GAUSTF achieves spectral accuracy comparable to the widely used FSDAF while maintaining high computational efficiency. Although it may not surpass state-of-the-art accuracy metrics in all scenarios, GAUSTF's primary contribution lies in its theoretical unification and mathematical transparency. It requires only a single pair of fine-resolution images and minimal computational resources. Consequently, GAUSTF offers a robust baseline for researchers with limited computing power and holds significant potential to serve as a differentiable physical constraint layer in future physics-informed deep learning models. The GAUSTF code is available at: https://doi.org/10.5281/zenodo.20373792**

***Index Terms*—Data fusion, Gaussian distribution, remote sensing, spatiotemporal fusion (STF), time-series.**

I. Introduction

THE analysis of multitemporal image sequences is essential and valuable in various remote sensing applications, such as vegetation/crop tracking [1], [2], evapotranspiration estimation [3], atmospheric monitoring [4], land-cover/land-use change detection [5], [6], and many other detailed examinations of human-nature interactions. Changes on the Earth's surface often occur rapidly and require both high temporal and spatial resolution for timely and accurate representation. However, high-resolution Earth observation (EO) data with both spatial and temporal continuity is limited due to there being insufficient remote sensing sensors and unstable observation conditions [7], [8]. For instance, commonly used open-access sensors with medium to high spatial resolution, such as the Landsat 8/9 constellation and Sentinel-2 MSI (Sentinel-2A/B/C), provide revisit intervals of 8–16 days at 30-m resolution and 5 days at 10–60-m resolution, respectively. In contrast, medium to low spatial resolution satellite instruments, including the legacy MODIS sensors and their modern successor VIIRS (375–750 m), as well as Sentinel-3 (300 m) and geostationary sensors (e.g., GOCI and Himawari), achieve much higher temporal resolutions of approximately 1–2 days. Moreover, cloud and cloud-shadow contamination significantly restrict the spatiotemporal availability of observations, which in turn hampers the practical applications of remote sensing data [9], [10], [11]. Fusing high spatial resolution data with high temporal resolution data combines their complementary characteristics, aiming to produce synthetic observations with enhanced spatial detail and temporal continuity.

To bridge this observational gap, spatiotemporal fusion (STF) algorithms have been developed to generate synthetic data with both high spatial and high temporal resolution. Fundamentally, STF aims to predict a fine-resolution image at an unobserved date t_2 by synergizing the spatial structural details from reference fine-resolution images at base time t_1 with the temporal evolutionary information extracted from coarse-resolution time-series. Over the past decades, research in this domain can be broadly categorized into two primary paradigms: data-driven methods and model-driven methods [12], [13], [14]. Data-driven approaches primarily utilize machine learning techniques—from sparse representation to advanced deep neural networks—to learn implicit nonlinear mappings between sensor modalities from large-scale training data. Conversely, model-driven methods reconstruct the target signal based on explicit physical or statistical priors, such as spectral mixing mechanisms or spatial dependence theories,

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formulating the fusion task as an inverse problem or a mathematical reconstruction process.

Model-driven methods constitute a foundational paradigm in STF and can be broadly categorized into process-based and statistical-based approaches. Process-based methods typically adhere to a strict operational sequence, distributing residuals by employing either spatially adaptive weighting or linear spectral unmixing strategies. The pioneering STARFM algorithm exemplifies the weighting strategy, which simplifies fusion by calculating weights based on spectral, temporal, and spatial differences within a moving window [15]. To address heterogeneous landscapes where STARFM struggles, improved algorithms such as ESTARFM, STAARCH, and RASTFM were developed to better capture complex land-cover and structural changes [16], [17]. Similarly, other process-based algorithms integrate unmixing techniques to resolve mixed pixel issues within the fusion workflow. A notable contribution in this domain came from STDFA, which shifted the focus to unmixing temporal changes [18]. This paradigm established a precedent for subsequent robust algorithms such as Fit-FC [14] and FSDAF [19], which combine unmixing with spatial interpolation. Alternatively, statistical-based methods frame fusion as an estimation problem, often utilizing a Bayesian framework to solve for maximum a posteriori (MAP) estimates. These methods establish relationships using point spread functions (PSFs) or joint Gaussian distributions, providing theoretical rigor to the fusion process [20], [21], [22], [23]. However, despite these advancements, most current model-driven methods still rely heavily on rigid empirical rules, requiring complex window searches and pixel-wise filtering [24], [25], [26], [27], [28], [29], [30], [31], [32]. Although effective as baselines, these traditional methods lack high-level mathematical abstraction. Because they allocate residuals based on local window searches, they generally fail to treat land-cover as coherent objects, making it difficult to utilize inherent physical properties like phenology. Moreover, their rigid, step-by-step processes may expose the models to cumulative error propagation, while their reliance on moving window searches tends to increase sensitivity to local data noise and heuristic hyperparameters.

Parallel to model-driven approaches, data-driven methods have evolved rapidly, shifting from traditional sparse representation to advanced deep learning architectures. Early attempts, such as sparse representation-based methods such as SPSTFM [33], introduced compressed sensing theory to STF, assuming that high- and low-resolution image patches share consistent sparse coefficients within learned over-complete dictionaries. While effective in recovering sharp edges, these methods are computationally intensive and highly sensitive to image misregistration [12], [34], [35], [36], [37], [38], [39]. To capture more complex relationships, traditional machine learning techniques were employed to model nonlinear mappings, yet they often struggled with high-dimensional feature extraction [40], [41]. The advent of deep learning revolutionized this domain, enabling the automatic learning of intricate nonlinear mappings from large-scale datasets. Convolutional neural networks (CNNs), such as DCSTFN [42] and HCNNet [43], first demonstrated the power of hierarchical feature extraction in preserving spatial details. Subsequently, generative adversarial networks (GANs) were introduced to alleviate the over-smoothing effects of CNNs by reconstructing realistic high-frequency textures via adversarial training [44], [45]. More recently, transformers and foundation models have emerged to capture long-range spatiotemporal dependencies and enhance generalization through self-supervised pretraining on massive multimodal datasets [46], [47], [48]. To further optimize this paradigm, efficient variants like EST-STFM employ sparse attention to reduce computational overhead [49], while state-of-the-art diffusion models (e.g., STFDiff) synthesize high-fidelity details via iterative denoising [50]. However, despite these strides in generative quality and efficiency, most data-driven models fundamentally operate as "black boxes" with limited interpretability. Their heavy reliance on fine-coarse training pairs severely limits generalizability when applied to new geographical regions or distinct phenological phases. While recent frameworks attempt to mitigate these issues through semantic priors [51], task decoupling [52], or by sequentially hybridizing neural networks with traditional algorithms [53], they inherently remain reliant on pretrained weights and rigid multistage pipelines. Because these approaches—including early attempts to embed physical constraints [54]—lack a statistically explicit formulation, they still struggle to guarantee physical consistency. Fundamentally, because traditional algorithms relied on rigid, step-by-step operations lacking mathematical abstraction, hybrid deep-learning models lack a physically rigorous framework to integrate. Overcoming this requires elevating the fusion process into a unified mathematical framework—one that ensures physical rigor and provides a transparent layer to naturally constrain data-driven models.

To address these compounded limitations, this study proposes a novel Gaussian model-based STF method (GAUSTF). GAUSTF innovatively bridges the physical mechanism of linear spectral unmixing with statistical inference by hypothesizing that the temporal reflectance evolution of each object follows a Gaussian distribution. Consequently, a coarse mixed pixel is no longer treated as a simple numerical average, but rather mathematically modeled as a linear combination of these distinct subpixel statistical distributions. By doing so, the complex fusion task is elegantly transformed into a rigorous parameter inference problem: estimating the mean and variance for each land-cover class. This object-level approach enables the model to naturally incorporate physical properties into the fusion process. Instead of searching for local pixel similarities, the inferred Gaussian parameters inherently capture the macroscopic physical trajectories and phenological consistencies of specific land covers as a coherent whole. In addition, GAUSTF achieves superior data efficiency grounded in this transparent mathematical structure.

II. Methodology

A. Notations and Definitions

Before describing the details of GAUSTF, some notations and definitions are given here for convenience.

1) y_{obs} is the coarse-resolution image pixel value, which is an observation.

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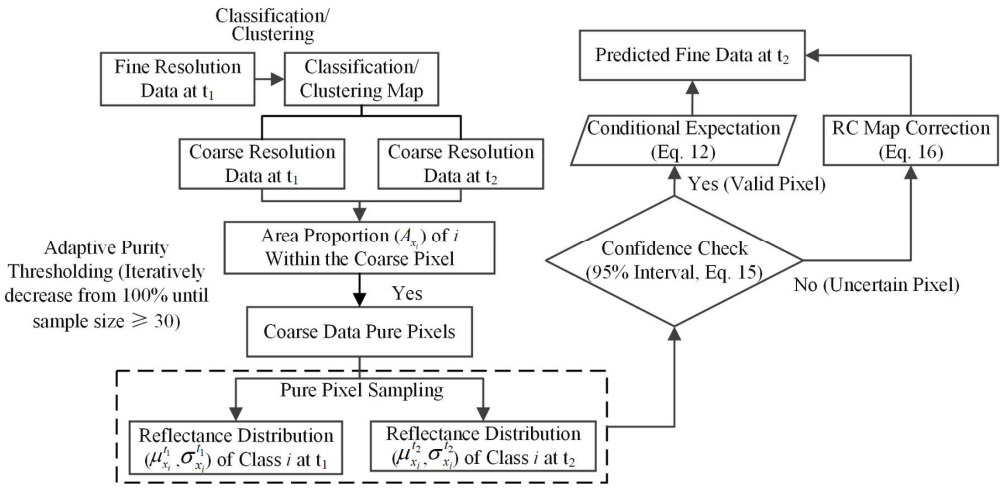


Fig. 1. Overall flowchart of the GAUSTF framework. The process is divided into three main stages: 1) extracting the area proportion (A_{x_i}) of land-covers using fine-resolution clustering; 2) adaptively sampling pure pixels to construct Gaussian distribution parameters (μ_{x_i}, σ_{x_i}) for both dates; and 3) predicting the target reflectance via conditional expectation, coupled with an uncertainty detection module that applies RC to pixels falling outside the 95% confidence interval.

2) y_{model} is the simulated coarse-resolution image pixel value presented as a random variable.
3) x_i is the class i's reflectance, presenting as random variables.
4) A_{x_i} is the class i's area proportion in the coarse-resolution image pixel, which is an observation.
5) μ_{x_i} is the mean of random variable x_i.
6) $\sigma^2_{x_i}$ is the variance of random variable x_i.
7) t_k represent the date k.

B. Theory Foundation

The GAUSTF framework fundamentally operates on the assumption that the temporal reflectance changes (RCs) of homogeneous land-cover objects approximate a Gaussian distribution. As illustrated in the detailed flowchart (Fig. 1), the workflow consists of three primary phases. First, fine-resolution data at the base date (t_1) is clustered to delineate object boundaries and calculate the area proportion of each land-cover class within the corresponding coarse pixels. Second, using an adaptive pure-pixel thresholding strategy, the algorithm extracts pure pixels from the coarse data to construct statistical Gaussian distributions (characterized by mean μ and variance σ^2) for each class at both t_1 and t_2. Finally, utilizing the principle of conditional expectation based on linear spectral unmixing, the method infers the fine-resolution reflectance at t_2. To ensure robustness against complex surface heterogeneity and abrupt phenological changes, GAUSTF incorporates a strict uncertainty modeling mechanism: pixels deviating from a 95% confidence interval are flagged as "uncertain" and are subsequently corrected using an RC strategy. This statistically explicit framework yields accurate predictions while preserving object-level spatial textures.

The following is the mathematical formulation of the theoretical foundation of GAUSTF.

According to the linear unmixing model

$$y_{model} = \sum_i A_{x_i} x_i + \varepsilon \tag{1}$$

where ε is the residual error. y_{model} is the simulated coarse-resolution image pixel value. A_{x_i} is the area proportion of class$i$. x_i is the spectral value component of class i.

Here, the x_i are treated as independent random variables, and we assume that they follow independent Gaussian distributions. The distribution of x_i is denoted as $N(\mu_{x_i}, \sigma^2_{x_i})$. Because a linear combination of independent Gaussian distributions results in another Gaussian distribution, the distribution of the linear combination $\sum A_{x_i} x_i$ is given by

$$y_{model} \sim N\left(\sum_i A_{x_i}\mu_{x_i}, \sum_i {A_{x_i}}^2 {\sigma_{x_i}}^2\right). \tag{2}$$

By determining the distribution for each type of land-cover, we can estimate the distribution of the mixed pixels using (2).

Due to factors such as the observation angle, varying aerosol conditions, and topographic influences, the actual observation of coarse pixel values y_{obs} with the same distribution can differ. However, determining the distribution and observed data provides insights into x_i. This is because we already know the distribution of x_i. If the variance of x_i is high, it suggests that the reflectance of class i is more susceptible to environmental or sensor observational variations. Consequently, greater discrepancies between the modeled y_{model} and the observed pixel value y_{obs} can be attributed to these factors. Statistically, this inference is supported by the conditional expectation, which is represented as follows:

$$\mathrm{E}\left[x_i|y\right] = \mu_{x_i} + \frac{Cov\left(x_i, y\right)}{{\sigma_y}^2}\left(y - \mu_y\right). \tag{3}$$

In this case, $Cov(x_i, y) = A_{x_i}\sigma^2_{x_i}$. Substituting into the above equation, we obtain

$$\mathrm{E}\left[x_i|y\right] = \mu_{x_i} + \frac{A_{x_i}{\sigma_{x_i}}^2}{{\sigma_y}^2}\left(y - \mu_y\right) \tag{4}$$

where we already get the σ^2_y and μ_y in (2), bring in (4)

$$x_i = \mu_{x_i} + \frac{A_{x_i}{\sigma_{x_i}}^2}{\sum_i {A_{x_i}}^2 {\sigma_{x_i}}^2}\left(y_{obs} - \sum_i A_{x_i}\mu_{x_i}\right) \tag{5}$$

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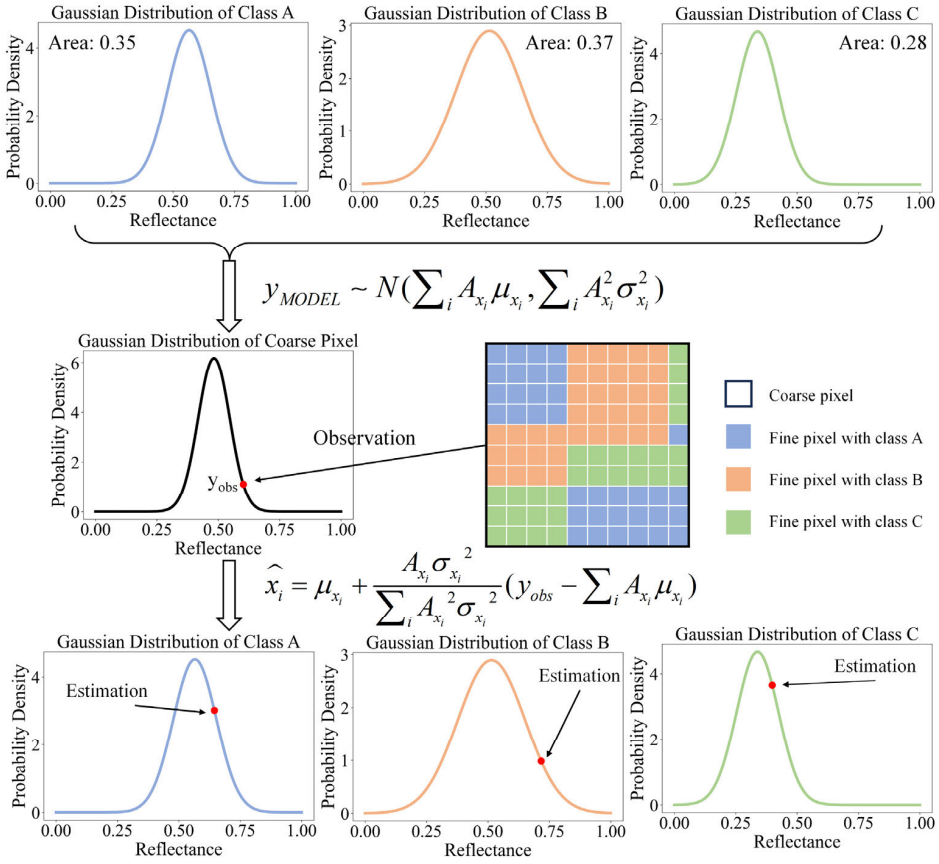


Fig. 2. Theory foundation of GAUSTF.

where $\hat{x}_i$ is the localized reflectance of type i. We can get the localized reflectance of type i at any given time t_k with coarse-resolution image observations. μ_{x_i} and σ_{x_i} can be obtained by sampling on coarse data on time k. Fig. 2 is an example.

Unlike previous Bayesian-based methods [55] and methods applying Bayesian principles like the spatial and temporal reflectance unmixing model (STRUM) [56], which typically assume that all coarse-resolution pixels share a global distribution, typically a joint multivariate Gaussian. GAUSTF models each coarse-resolution pixel using an independent Gaussian distribution. While these pixels are fundamentally instances of a joint Gaussian distribution, treating them independently drastically simplifies the parameter inference process.

C. Theory Derivation

While the preceding theory is applicable to mixed-pixel unmixing, its direct implementation often yields suboptimal results. This is primarily because raw surface reflectance deviates from a strict Gaussian distribution, compounded by inherent spectral discrepancies between coarse and fine-resolution imagery [57]. To overcome these constraints, performance can be remarkably improved by modeling temporal RCs rather than absolute reflectance. Statistically, supported by the convolution theorem and the dilution of higher order cumulants [57], the subtraction of two Gaussian-like variables inherently forces the resulting distribution to approximate a stricter Gaussian form. Complementing this mathematical rigor, classical model-based methods like STARFM have empirically revealed that coarse and fine data share consistent temporal spectral differences. Driven by both statistical theorems and remote sensing empiricism, the application of (6) and its derivation [up to (12)] are substantiated. The derivation of the GAUSTF method proceeds as follows.

Under the assumption that t_1 and t_2 share the same error term, we obtain

$$y_{model}^{t_2} - y_{model}^{t_1} = \sum_i A_{x_i}^{t_2} x_i^{t_2} - \sum_i A_{x_i}^{t_1} x_i^{t_1}. \tag{6}$$

Given that the difference between two independent normally distributed random variables remains normally distributed, $y_{model}^{t_2} - y_{model}^{t_1}$ also follows a Gaussian distribution. According to the linear combination properties of independent Gaussian variables, its distribution is formulated as

$$\begin{aligned} y_{model}^{t_2} - y_{model}^{t_1} \sim N\Bigg(& \sum_i A_{x_i}^{t_2}\mu_{x_i}^{t_2} - \sum_i A_{x_i}^{t_1}\mu_{x_i}^{t_1}, \sum_i \left(A_{x_i}^{t_2}\right)^2 \left(\sigma_{x_i}^{t_2}\right)^2 \\ & + \sum_i \left(A_{x_i}^{t_1}\right)^2 \left(\sigma_{x_i}^{t_1}\right)^2 \Bigg). \end{aligned} \tag{7}$$

Building upon the conditional expectation framework introduced in (3), we formulate the conditional expectation for the temporal RC, $x_i^{t_2} - x_i^{t_1}$ as follows:

$$\begin{aligned} & \mathrm{E}\left[x_i^{t_2} - x_i^{t_1} | y^{t_2} - y^{t_1}\right] \\ & = \mu_{x_i^{t_2} - x_i^{t_1}} + \frac{Cov\left(x_i^{t_2} - x_i^{t_1}, y^{t_2} - y^{t_1}\right)}{\sigma^2_{y^{t_2} - y^{t_1}}} \left(y^{t_2} - y^{t_1} - \mu_{y^{t_2} - y^{t_1}}\right). \end{aligned} \tag{8}$$

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Furthermore, given the covariance relationship $Cov(x_i^{t_2} - x_i^{t_1}, y^{t_2} - y^{t_1}) = A_{x_i}^{t_2} \times {\sigma_{x_i}^{t_2}}^2 + A_{x_i}^{t_1} \times {\sigma_{x_i}^{t_1}}^2$, substituting this term into (8) yields

$$\begin{aligned} &\mathrm{E}\left[x_i^{t_2} - x_i^{t_1} \middle| y^{t_2} - y^{t_1}\right] \\ &= \mu_{x_i^{t_2} - x_i^{t_1}} + \frac{A_{x_i}^{t_2} {\sigma_{x_i}^{t_2}}^2 + A_{x_i}^{t_1} {\sigma_{x_i}^{t_1}}^2}{{\sigma_{y^{t_2} - y^{t_1}}}^2} \left(y^{t_2} - y^{t_1} - \mu_{y^{t_2} - y^{t_1}}\right). \end{aligned} \tag{9}$$

Analogous to the derivation of (5), the optimal estimation of the temporal change is achieved by substituting the actual coarse observation ($y_{obs}^{t_1}$ and $y_{obs}^{t_2}$) and the mean difference into the equation

$$\begin{aligned} x_i^{t_2} - x_i^{t_1} = \mu_{x_i^{t_2} - x_i^{t_1}} + \frac{A_{x_i}^{t_2} \times {\sigma_{x_i}^{t_2}}^2 + A_{x_i}^{t_1} {\sigma_{x_i}^{t_1}}^2}{\sum_i A_{x_i}^{t_2} {\sigma_{x_i}^{t_2}}^2 + \sum_i A_{x_i}^{t_1} {\sigma_{x_i}^{t_1}}^2} \\ \left(y_{obs}^{t_2} - y_{obs}^{t_1} - \left(\sum_i A_{x_i}^{t_2} \mu_{x_i}^{t_2} - \sum_i A_{x_i}^{t_1} \mu_{x_i}^{t_1}\right)\right) \end{aligned} \tag{10}$$

where $y_{obs}^{t_1}$ and $y_{obs}^{t_2}$ denote the observed coarse pixel values at t_1 and t_2, respectively. Since the fine-resolution variables $x_i^{t_1}$ and $x_i^{t_2}$ inherently follow Gaussian distributions, the expectation of their difference is exactly the difference of their expectations, i.e., $\mu_{x_i^{t_2} - x_i^{t_1}} = \mu_{x_i^{t_2}} - \mu_{x_i^{t_1}}$. Applying this property, we obtain the final formulation

$$\begin{aligned} x_i^{t_2} - x_i^{t_1} = \mu_{x_i}^{t_2} - \mu_{x_i}^{t_1} + \frac{A_{x_i}^{t_2} \times {\sigma_{x_i}^{t_2}}^2 + A_{x_i}^{t_1} {\sigma_{x_i}^{t_1}}^2}{\sum_i A_{x_i}^{t_2} {\sigma_{x_i}^{t_2}}^2 + \sum_i A_{x_i}^{t_1} {\sigma_{x_i}^{t_1}}^2} \\ \left(y_{obs}^{t_2} - y_{obs}^{t_1} - \left(\sum_i A_{x_i}^{t_2} \mu_{x_i}^{t_2} - \sum_i A_{x_i}^{t_1} \mu_{x_i}^{t_1}\right)\right). \end{aligned} \tag{11}$$

To simplify the notation, we denote the right side of (11) as the RC (RC_{x_i}), which represents the pixel-wise variation in surface reflectance. Since $x_i^{t_1}$ is a known value obtained from fine-resolution data at t_1, a preliminary estimation is given by $\widehat{x_i^{t_2}} = x_i^{t_1} + RC_{x_i}$, where temporal change and local variation are both considered in the algorithm.

Under the assumption of invariant land-cover, $A_{x_i}^{t_1} = A_{x_i}^{t_2}$, and the equation can be formulated as

$$\begin{aligned} x_i^{t_2} - x_i^{t_1} = \mu_{x_i}^{t_2} - \mu_{x_i}^{t_1} + \frac{A_{x_i} \times \left({\sigma_{x_i}^{t_2}}^2 + {\sigma_{x_i}^{t_1}}^2\right)}{\sum_i {A_{x_i}}^2 {\sigma_{x_i}^{t_2}}^2 + \sum_i {A_{x_i}}^2 {\sigma_{x_i}^{t_1}}^2} \\ \left(y_{obs}^{t_2} - y_{obs}^{t_1} - \left(\sum_i A_{x_i} \mu_{x_i}^{t_2} - \sum_i A_{x_i} \mu_{x_i}^{t_1}\right)\right) \end{aligned} \tag{12}$$

where A_{x_i} is the area proportion of land cover x_i within a coarse pixel.

D. Uncertainty Modeling of Land-Covers

While the Gaussian distribution effectively approximates the reflectance of most land covers, deviations may occur due to complex surface heterogeneity or abrupt changes. To ensure resilient estimation, we employ a 95% confidence level threshold to distinguish reliable predictions from uncertain ones. Crucially, this threshold serves a dual purpose: it not only identifies intrinsic model uncertainty (where the Gaussian assumption may not hold) but also acts as a detector for abrupt land-cover changes. According to the standard normal distribution, the Z-value for a 95% confidence level is 1.96. Based on (2), the validity check is formulated as

$$y_{model} \sim N\left(\sum_i A_{x_i} \mu_{x_i}, \sum_i {A_{x_i}}^2 {\sigma_{x_i}}^2\right). \tag{13}$$

If a pixel remains unchanged in land-cover type and strictly adheres to the Gaussian assumption, it should theoretically fall within the 95% confidence interval defined in the following equation:

$$P\left(-1.96 \leq \frac{X - \mu}{\sigma} \leq 1.96\right) = 0.95. \tag{14}$$

Consequently, the valid range for the predicted value is derived as shown in the following equation:

$$\begin{aligned} Uncertain_{x_i} \notin \left[\sum_i A_{x_i} \mu_{x_i}^{t_2} - 1.96^* \sqrt{\sum_i {A_{x_i}}^2 {\sigma_{x_i}^{t_2}}^2}, \sum_i A_{x_i} \mu_{x_i}^{t_2} \right. \\ \left. + 1.96^* \sqrt{\sum_i {A_{x_i}}^2 {\sigma_{x_i}^{t_2}}^2}\right]. \end{aligned} \tag{15}$$

Importantly, confining the predictions to this 95% interval mathematically enforces our core assumption, as data within this range exhibit a significantly stricter approximation to the Gaussian distribution. We have empirically verified this phenomenon using Landsat and Sentinel observations across our study sites, with the detailed statistical validations provided in the Supplementary Material B.

Conversely, pixels falling outside this valid range are flagged as "uncertain pixels." This exclusion mechanism is critical because it captures two types of anomalies: 1) pixels where the actual spectral distribution deviates significantly from the Gaussian assumption and 2) pixels that have undergone land-cover change between t_1 and t_2, thereby violating the statistical parameters derived from the base date. For these identified uncertain pixels, we employ the following equation to estimate the RC map, leveraging the coarse-resolution information to correct the deviation:

$$x_i^{t_2} - x_i^{t_1} = y_{obs}^{t_2} - y_{obs}^{t_1}. \tag{16}$$

To further refine the estimation of $x_i^{t_2}$, we incorporate fine-resolution spatial information into the uncertain pixels using the following strategy: $\widehat{x_i^{t_2}} = x_i^{t_1} + y_{obs}^{t_2} - y_{obs}^{t_1}$. This operation is conducted for each band individually. It is worth noting that uncertainty assessment is intrinsically band-specific. For instance, a pixel may be flagged as uncertain in the NIR band (e.g., due to vegetation change) but remain reliable in the Red band. In such cases, the algorithm adaptively applies the RC map correction strictly to the specific bands where the confidence test fails, while relying on the standard model for the changed bands.

E. GAUSTF Implementation

We find that the results can be improved by utilizing interpolated coarse images, as this approach captures finer

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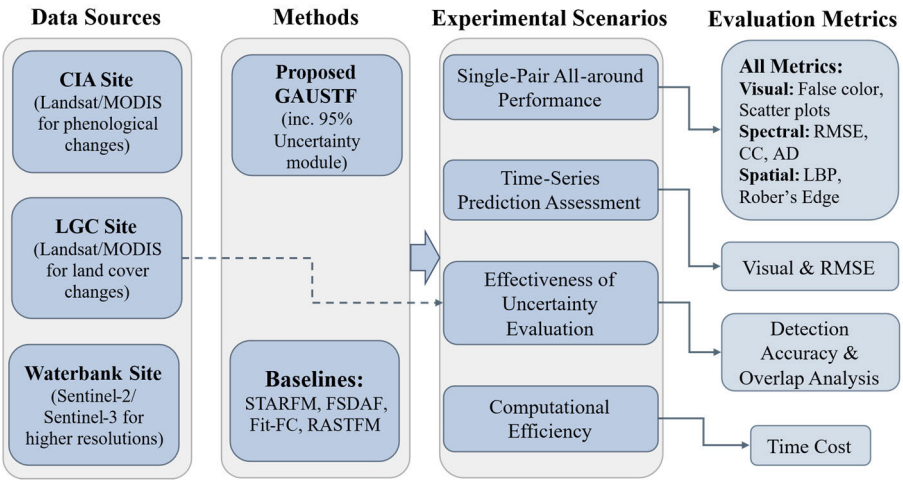


Fig. 3. Schematic workflow of the experimental design and evaluation framework.

spatial details and effectively mitigates block effects. In other words, $y_{obs}^{t_2}$ and $y_{obs}^{t_1}$ are obtained using the interpolated coarse images at time t_2 and t_1. In this study, we employ cubic interpolation to resample the coarse data to the fine resolution.

To implement GAUSTF, we require clustering maps at t_1 or t_2. These maps allow us to sample the image to obtain values for $\mu_{x_i}^{t_1}\mu_{x_i}^{t_2}$, $A_{x_i}^{t_1}, A_{x_i}^{t_2}\sigma_{x_i}^{t_1}$, and $\sigma_{x_i}^{t_2}$. More specifically, in this study, $\mu_{x_i}^{t_1}$ and $\sigma_{x_i}^{t_1}$ are derived by sampling pure pixels from the coarse data at t_1, while $A_{x_i}^{t_1}$ is obtained by calculating the area proportion of class i within a coarse data pixel at t_1. Under the assumption of invariant land-cover ($A_{x_i}^{t_1} = A_{x_i}^{t_2}$), we can simply sample pure pixels from the coarse data at t_2 using the clustering map generated at t_1.

For the parameterization of the Gaussian distribution, two key variables are crucial: the number of clusters (classes) and the pure pixel threshold. The number of classes must strike a delicate balance to ensure that the extracted signal closely approximates a Gaussian distribution. If the number of classes is too small, highly mixed land covers will fail to conform to a Gaussian assumption. Conversely, over-segmentation may result in an insufficient number of pure pixels, hindering the accurate statistical modeling of specific classes. Pure pixels in the coarse data at t_2 are identified by searching within the spatial boundaries of each coarse pixel for areas dominated by a single land-cover class. However, in heterogeneous landscapes, absolutely pure pixels may be exceedingly scarce, leading to insufficient samples for stable Gaussian modeling (according to the central limit theorem (CLT), it is generally recommended to have no fewer than 30 samples). Therefore, defining an appropriate pure pixel threshold—the required proportion of fine-resolution pixels belonging to a target class within a coarse pixel—is crucial. A threshold set too low introduces impurities, whereas a threshold set too high yields an inadequate sample size (we will discuss this in detail in Section V-C).

To resolve this paradox, we developed an automated, adaptive thresholding strategy within our algorithm. Initially, the algorithm seeks perfectly pure pixels (a 100% threshold). If the number of identified pure pixels for a given class is insufficient (fewer than 30), the algorithm iteratively relaxes the purity threshold by 5% decrements (e.g., to 95%, 90%) until a statistically significant sample size (≥30 pure pixels) is successfully accumulated.

GAUSTF is utilized in conjunction with a fine-resolution image, a clustering or classification map derived from this fine image, and two coarse-resolution images. In practice, as long as we have data on the distribution of land-cover reflectance at two time points (t_1 and t_2), the signal values at all locations for t_2 can be accurately predicted. This endows the algorithm with flexibility and scalability. For example, μ_{x_i} can be temporally smoothed using the Savitzky–Golay filter, given that the reflectance of a certain landscape is generally continuous over time. Furthermore, σ_{x_i} can be estimated based on prior knowledge. For instance, as the temporal gap increases, the variance of class reflectance tends to increase during pure pixel sampling. Endmember extraction techniques can also be employed to constrain this variance.

III. Testing Experiment

To comprehensively evaluate the performance of the proposed GAUSTF algorithm in complex STF tasks, we designed a multidimensional experimental framework. As illustrated in the schematic workflow (Fig. 3), our evaluation spans across different data sources (Coleambally irrigation area (CIA), lower Gwydir catchment (LGC), and Waterbank sites) and encompasses four key experimental scenarios: all-around performance, time-series prediction, uncertainty mechanism ablation, and computational efficiency. Each scenario is assessed using a targeted set of visual, spectral, and spatial metrics to ensure a stable and objective validation of the model. Although comprehensive evaluations were conducted for a broader set of benchmark models, for the sake of clarity and conciseness, the main manuscript only presents the comparative results of FSDAF and STARFM against our proposed GAUSTF. Exhaustive quantitative and qualitative evaluations for all methods (i.e., FSDAF, STARFM, Fit-FC, RASTFM, and GAUSTF) in the one-pair fusion task are detailed in Supplementary Material C. Furthermore, to facilitate reproducibility and future research, the source code

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for all methods utilized in this study has been packaged and made publicly available at: https://github.com/Guanxudong-code/STF-Comparison-Framework

A. Study Area and Data

Three datasets are used for method comparison. The first two are Landsat-MODIS pairs provided by Emelyanova et al. [58], which are commonly used as benchmarks by the community. The third is a benchmark dataset consisting of Sentinel-2 (S2) and Sentinel-3 (S3) imagery, which has a larger resolution difference, with the fine data offering a higher resolution [59].

The first site is the CIA, located in southern New South Wales (NSW, Australia; 145.0675°E, 34.0034°S). This dataset is commonly used to test the effectiveness of STF methods in predicting fine-resolution data in heterogeneous landscapes with drastic phenological changes. There are 16 pairs of Landsat 7 ETM+ and MOD09GA images obtained from October 2001 to May 2002, covering an area of approximately 20×20 km (800×800 pixels, 25-m spatial resolution). The CIA site has a heterogeneous landscape consisting of many small parcels of rice croplands and woodlands. Significant temporal changes have occurred in this area, including phenological changes of the vegetation and land-cover changes caused by agricultural activities.

The second site is the LGC, located in northern New South Wales, Australia (149.2815°E, 29.0855°S). This dataset is commonly used to test the effectiveness of STF methods in predicting fine-resolution data during land-cover change scenarios. There are 14 pairs of Landsat 7 ETM+ and MOD09GA images obtained from April 2004 to April 2005. This site covers an area of 20×20 km and is relatively homogeneous compared to the first site, with large parcels of croplands and natural vegetation. A significant flood occurred in December 2004, causing land-cover type changes to water in some pixels.

The third dataset is Sentinel-2 (S2) and Sentinel-3 (S3) pairs provided by Boumahdi et al. [59]; we used the Waterbank dataset within the dataset. The dataset uses S2 level 1C and S3 OLCI level 1B data. S2 and S3 data were atmospherically corrected and reprojected for S2 and S3 projection matching. The Waterbank site, located in Waterbank (122.7259°E, 17.706°S), is a rural site in northern Australia. It exhibits rapid temporal changes due to forest fires. This location is well known for the high frequency of early season fires, which leads to different phenological changes in vegetation throughout the year. The data cover an area of 18×18 km (1800×1800 Sentinel-2 pixels). In our study, we use S2 band 2 (Blue band), S2 band 3 (Green band), S2 band 4 (Red band), and S2 band 8 (NIR band) to fuse S3 Oa4 band, S3 Oa6 band, S3 Oa8 band, and S3 Oa17 band. These correspond to the 1, 2, 3, and 7 bands of S2 data and bands 4, 6, 8, and 13 of the S3 data.

Prior to the execution of the fusion algorithms, a basic data quality control step was applied uniformly to all three datasets. Specifically, we automatically masked nonphysical anomalous reflectance values falling outside the valid range of [0, 10 000]. This outlier removal process aims to eliminate extreme data points typically caused by sensor errors or severe atmospheric artifacts, thereby ensuring the stability of the input data for the subsequent normal distribution assumption and uncertainty analysis.

B. Comparison and Evaluation Metrics

Because the method only uses one fine dataset, we did not select learning-based methods for comparison, as they require a time series of fine data. Instead, we compared our method with the STARFM and FSDAF methods, which also require only one fine dataset. Visual comparisons of false color composites and actual-predicted images were conducted. Scatter plots were used for a qualitative evaluation of fusion results. Four suggested metrics [60] and the correlation coefficient (CC or r) were used to quantitatively assess the performance. The root-mean-square error (RMSE) was used to measure the difference between the predicted reflectance and the actual reflectance. The CC was used to show the linear relationship between predicted and actual reflectance. The absolute difference (AD) is the average of all absolute errors between the predicted and actual pixel values

$$AD = \frac{1}{n}\sum_{i=1}^{n}\left(x_{predicted} - x_{actual}\right). \tag{17}$$

In addition to the above quantitative assessments, which place emphasis on spectral similarity, local binary patterns (LBPs) and Roberts' Edge are implemented for spatial similarity tests. The following is the equation for Roberts' Edge detection:

$$Edge = \left|D_{i,j} - D_{i+1,j+1}\right| + \left|D_{i,j+1} - D_{i+1,j}\right| \tag{18}$$

where $D_{i,j}$ is the value of pixels at ith row and jth column. The value of Edge is between -1 and 1; value 0 of Edge represents the perfect fused image; more negative values indicate edge features in the fused image over-smoothed; more positive values indicate edge features in the fused image over-sharpened. The following is the equation for LBP:

$$LBP = decimal\,(d1d2\cdots d8)$$
$$di = \begin{cases} 1, & \text{if Di} > \text{DC} \\ 0, & otherwise \end{cases} \tag{19}$$

where Di is the value of pixels surrounding the central pixel in a 3×3 moving window; DC is the value of the central pixel. Decimal is the operation to convert the binary digits to a decimal number. The value of LBP is between -1 and 1. Value 0 represents the perfect fused image; more negative values indicate textural features in the fused image over-smoothed; more positive values indicate textural features in the fused image over-sharpened.

IV. Results

The hardware configuration used in this study includes an Intel[1] Core[2] i9-10850K CPU @ 3.60 GHz, 64.0 GB DDR4 RAM, and a 1-TB NVMe SSD. We used Python 3.10 as the interpreter. It should be noted that in this study, Fit-FC and RASTFM primarily serve as baselines for computational efficiency (execution time) comparisons.

[1]Registered trademark.
[2]Trademarked.

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TABLE I

ACCURACY ASSESSMENT OF THREE DATA FUSION METHODS APPLIED TO THE STUDY SITE CIA, PREDICTION OF TWO DATES USING THE DATE NOVEMBER 25, 2001. THE UNITS ARE REFLECTANCE

Prediction of date 12 Jan 2002 using the date 25 Nov 2001															
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	0.7493	28.59	0.0189	-0.4320	**-0.0572**	**0.7834**	28.47	**0.0177**	-0.4176	-0.1564	0.7826	**27.89**	0.0177	**-0.3919**	-0.0970
Green	0.7498	27.96	0.0284	-0.4655	**-0.0688**	**0.7949**	28.00	**0.0261**	-0.4367	-0.0893	0.7884	**26.59**	0.0264	**-0.4283**	-0.0803
Red	0.7996	30.34	0.0450	-0.4444	-0.0912	**0.8367**	29.79	**0.0413**	-0.4128	**-0.0241**	0.8308	**27.87**	0.0416	**-0.3992**	-0.0261
NIR	0.4388	-15.43	0.0742	-0.3922	-0.0325	**0.5564**	**-14.91**	**0.0624**	-0.4628	**-0.0056**	0.5064	-16.18	0.0681	-0.4055	0.0060
SWIR1	0.8571	**-61.53**	0.0562	**-0.2117**	**0.0340**	**0.8756**	-63.57	**0.0508**	-0.2812	0.0483	0.8671	-63.36	0.0533	-0.2697	0.0504
SWIR2	0.8721	**-19.50**	0.0439	**-0.2194**	**0.0460**	**0.8791**	-23.61	**0.0421**	-0.2780	0.0639	0.8784	-20.17	0.0429	-0.2441	0.0754
absmean	0.7445	30.56	0.0444	0.3609	**0.0550**	**0.7877**	31.39	**0.0401**	0.3815	0.0646	0.7756	**30.34**	0.0417	**0.3565**	0.0559
Prediction of date 10 Mar 2002 using the date 25 Nov 2001															
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	**0.7552**	75.61	0.0163	-0.3426	0.0598	0.7527	**74.20**	**0.0163**	-0.3813	-0.0672	0.7327	76.17	0.0173	**-0.2964**	**0.0382**
Green	0.6958	43.57	0.0205	-0.3180	0.0889	**0.7109**	**42.08**	**0.0195**	-0.3825	**0.0093**	0.6834	45.10	0.0211	**-0.3112**	0.0803
Red	0.7726	21.25	0.0315	-0.2848	-0.0456	**0.7878**	**18.79**	**0.0298**	-0.3386	-0.0191	0.7713	22.44	0.0320	**-0.2748**	**0.0093**
NIR	0.0034	**15.89**	0.1130	**-0.3577**	**0.0531**	**0.3153**	18.17	**0.0893**	-0.4828	0.0905	0.2273	22.93	0.0962	-0.3879	0.1030
SWIR1	0.8249	-20.89	0.0607	**-0.2076**	-0.0341	**0.8421**	-27.30	**0.0536**	-0.2988	-0.0210	0.8308	**-18.94**	0.0578	-0.2652	**-0.0172**
SWIR2	0.8081	-13.11	0.0553	**-0.2764**	-0.0628	**0.8260**	-21.59	**0.0515**	-0.3459	-0.0407	0.8117	**-11.86**	0.0549	-0.3006	**-0.0314**
absmean	0.6433	**31.72**	0.0496	**0.2979**	0.0574	**0.7058**	33.69	**0.0433**	0.3716	**0.0413**	0.6762	32.91	0.0466	0.3060	0.0466

RMSE = Root Mean Square Error, CC = correlation coefficient, AD = average difference from true reflectance, Edge = Roberts' Edge, LBP = local binary patterns, absmean = absolute mean value

A. Single Pairs All-Around Performance in CIA

We applied K-means clustering to generate the clustering map, classifying the entire image into three classes in the CIA dataset. The ISODATA method from the original FSDAF implementation was used for its prediction (with the minimum and maximum number of classes set to 5 and 6, respectively). For FSDAF, the number of pure pixels was set to 360, the block size to 30, and both the window size and the number of similar pixels were set to 20.

We selected the image from November 25, 2001 to predict two subsequent dates: January 12, 2002 and March 10, 2002. The data were classified into three spectral classes based on the November 25, 2001 image. The MODIS data used in GAUSTF were resampled to 500 m, corresponding to the native 250–500-m spatial resolution of raw MODIS bands. Because some of the original MODIS data were not fully registered, we directly fed the bilinearly resampled data into GAUSTF, which requires the coarse data at its original resolution (500 m for the CIA dataset). For STARFM, which requires absolute alignment among the MODIS data pairs, the majority (mode) method in Python was applied to resample the 500-m MODIS data to a 25-m resolution. The original 25-m resampled MODIS data in the CIA dataset were directly fed into FSDAF.

Quantitative accuracy assessments of the three STF methods are presented in Table I. Overall, FSDAF achieved the best RMSE and CC across almost all bands on both prediction dates. We found that GAUSTF yielded a CC and RMSE close to those of FSDAF for the nearer prediction date (January 12, 2002). Furthermore, GAUSTF achieved the best AD, indicating a spectral prediction capability competitive with FSDAF and superior to STARFM. The scatter plot for the NIR band predictions is shown in Fig. 4, with the all-band scatter plots provided in the Supplementary Material A (Figs. SA1 and SA2). The plot reveals that FSDAF exhibits a tighter convergence in the NIR band prediction. GAUSTF produced an NIR band RMSE of 0.0681, outperforming STARFM; however, its ability to predict heterogeneous regions with acute reflectance variation remains weaker than that of FSDAF (RMSE is 0.0624). The NIR band scatter plots for the second date are shown in Fig. 4(d)–(f). FSDAF results are most closely aligned with the 1:1 line, and while GAUSTF and STARFM exhibit similar patterns, GAUSTF delivered better RMSE and CC values than STARFM. Nonetheless, FSDAF generated the best overall RMSE and CC. Notably, GAUSTF performed worse than STARFM in the first three bands—where the RMSE of Δy is relatively smaller, thereby giving STARFM an advantage (Table I)—but outperformed STARFM in the last three bands. FSDAF proved to be the most effective among the three methods, demonstrating the strongest spectral prediction capabilities.

However, FSDAF did not achieve the best performance across all indices, falling short in AD, Edge, and LBP. In addition, the difference in the mean RMSE between FSDAF and GAUSTF is marginal. Users may prefer GAUSTF for its well-balanced performance and computational efficiency, whereas FSDAF may be favored when superior spectral

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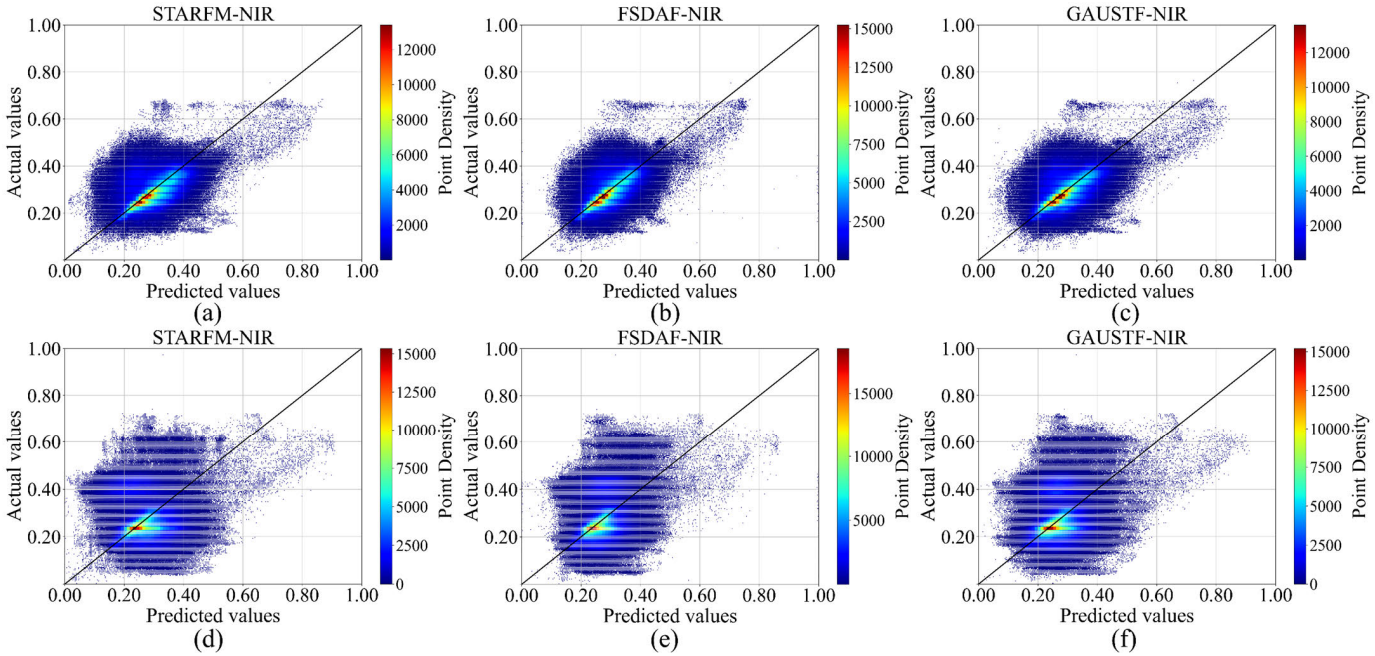


Fig. 4. Scatter plot of the NIR band of the three methods. (a)–(c) Prediction on January 12, 2002. (d)–(f) Prediction on March 10, 2002.

TABLE II
TEMPORAL AND CROSS-SENSOR STATISTICAL DIFFERENCES IN REFLECTANCE

			12 Jan 2002		10 Mar 2002	
	Diff(X^{t_1}, Y^{t_1})		Diff(Δy)		Diff(Δy)	
	RMSE	CC	RMSE	CC	RMSE	CC
Blue	0.0319	0.2903	0.0168	0.6350	0.0105	0.1398
Green	0.0323	0.3621	0.0276	0.6065	0.0130	0.1582
Red	0.0320	0.5745	0.0399	0.7021	0.0192	0.3182
NIR	0.0320	0.6561	0.0762	0.2110	0.0679	-0.1092
SWIR1	0.0475	0.8386	0.0601	0.8816	0.0386	0.6631
SWIR2	0.0416	0.8002	0.0384	0.8995	0.0379	0.6187

Diff(Δy) is the difference of reflectance between MODIS data at t_1 and t_2, Diff(X^{t_1},Y^{t_1}) is the difference of reflectance between MODIS data at t_1 and downscale resampling Landsat data at t_1.

prediction accuracy is paramount, despite its slower processing speed.

We find that two key factors influence the final prediction result of all the methods: the difference in reflectance over time and the difference between different sensors. These differences are illustrated in Table II.

The differences between the MODIS data at the two dates reflect the temporal variation in surface reflectance. As shown in Table II, the NIR band exhibits a substantially lower CC and a higher RMSE relative to the other bands, indicating pronounced phenological changes between the base date and the prediction date. Furthermore, while the intersensor differences for the NIR band are not substantial, all three methods demonstrate diminished predictive performance in this band. This suggests that the temporal change component, denoted as Δy, exerts a more profound influence on the final prediction than the discrepancies between sensors. This outcome is theoretically sound, as the temporal modeling phases of all three algorithms primarily rely on Δy rather than y^{t_2} to drive the reflectance prediction.

A detailed visual comparison of the algorithms' predictions for January 12, 2002 (t_2) across highly heterogeneous regions is presented in Fig. 5. The close-up views reveal pronounced differences in the reconstruction of spatial details. Specifically, STARFM suffers from a noticeable block effect. While FSDAF produces visually pleasing results, it tends to yield over-smoothed predictions, which is quantitatively corroborated by Table I, where its performance in the LBP and Edge metrics is relatively weaker. In contrast, GAUSTF maintains highly competitive performance on the LBP and Edge indices, demonstrating superior preservation of fine structural features. This advantage stems from GAUSTF's ability to generate a smooth and statistically convincing prediction for the RC map [as defined in (11) and (16)]. Consequently, by adding this refined RC map to the base Landsat data at t_1, GAUSTF successfully preserves the original high-resolution edge features of land-covers without introducing the blurring artifacts characteristic of FSDAF.

Fig. 6 presents a visual comparison of the three STF methods for the prediction date of March 10, 2002. A detailed examination of the highlighted subareas reveals that GAUSTF delivers a more consistent prediction for transient, object-level changes. In Subarea 1 (indicated by the upper yellow rectangle), a highly reflective object in the base image (f) transitions into soil and growing crops by the prediction date (g). Although all three methods attempt to capture this dynamic, STARFM (h) introduces a pronounced dark block artifact. While FSDAF (i) and GAUSTF (j) both achieve smoother transitions, GAUSTF generates a prediction that exhibits superior fidelity to the surrounding spatial texture. In Subarea 2 (within the yellow oval), the base image (k) features a prominent bright object that entirely disappears by the

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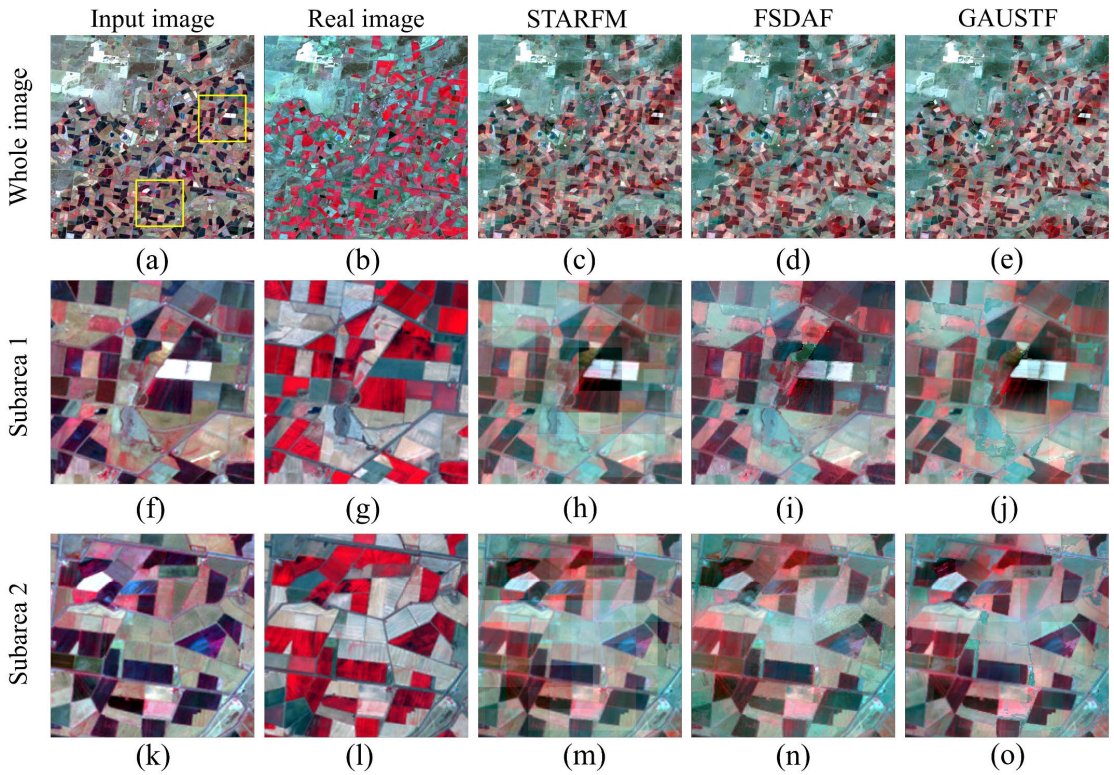


Fig. 5. Original Landsat images of the heterogeneous site acquired at t_1 and t_2 are (a) and (b), respectively; the predictions for January 12, 2002 by STARFM, FSDAF, and GAUSTF are shown in (c)–(e). (f)–(j) show the input image, real image, and predictions by STARFM, FSDAF, and GAUSTF for Subarea 1, respectively. (k)–(o) Show the corresponding detailed images for Subarea 2. Subarea 1 corresponds to the upper yellow rectangle of the whole image. Subarea 2 corresponds to the lower yellow rectangle of the whole image.

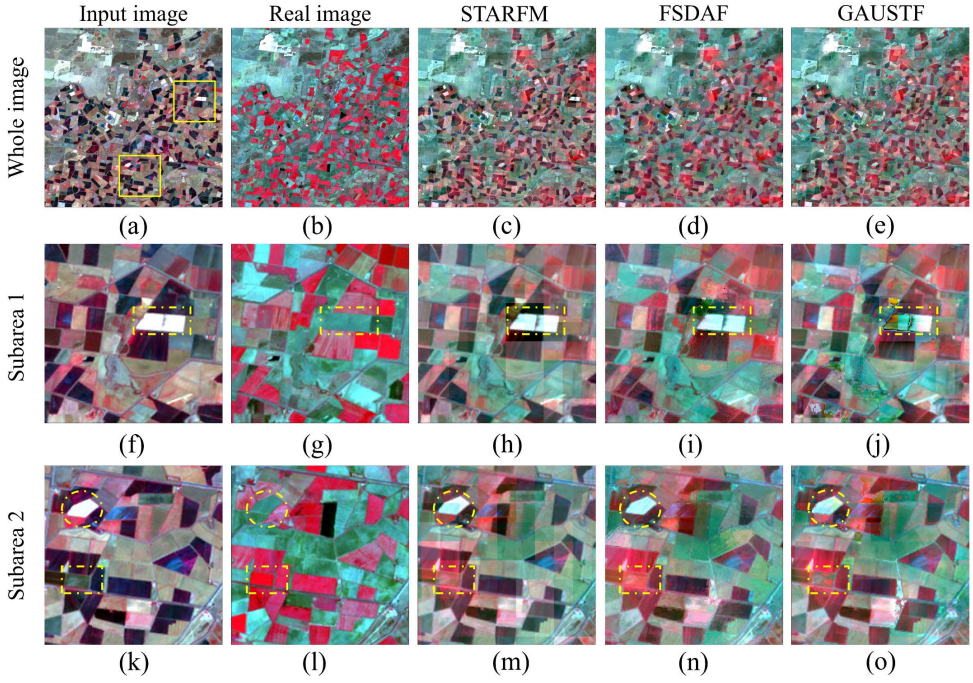


Fig. 6. Original Landsat images of the heterogeneous site acquired at t_1 and t_2 are (a) and (b); the predictions for March 10, 2002 by STARFM, FSDAF, and GAUSTF are shown in (c)–(e). (f)–(j) Show the input image, real image, and predictions by STARFM, FSDAF, and GAUSTF for Subarea 1, respectively. (k)–(o) show the corresponding detailed images for Subarea 2. Subarea 1 is the upper yellow rectangle of the whole image. Subarea 2 is the lower yellow rectangle of the whole image.

prediction date (l). STARFM (m) and FSDAF (n) fail to completely resolve this change, leaving residual spectral artifacts of the former object. Conversely, GAUSTF (o) successfully detects this structural disappearance, accurately predicting the spectral transition to match the adjacent land. Within the yellow rectangle of Subarea 2, all three methods effectively

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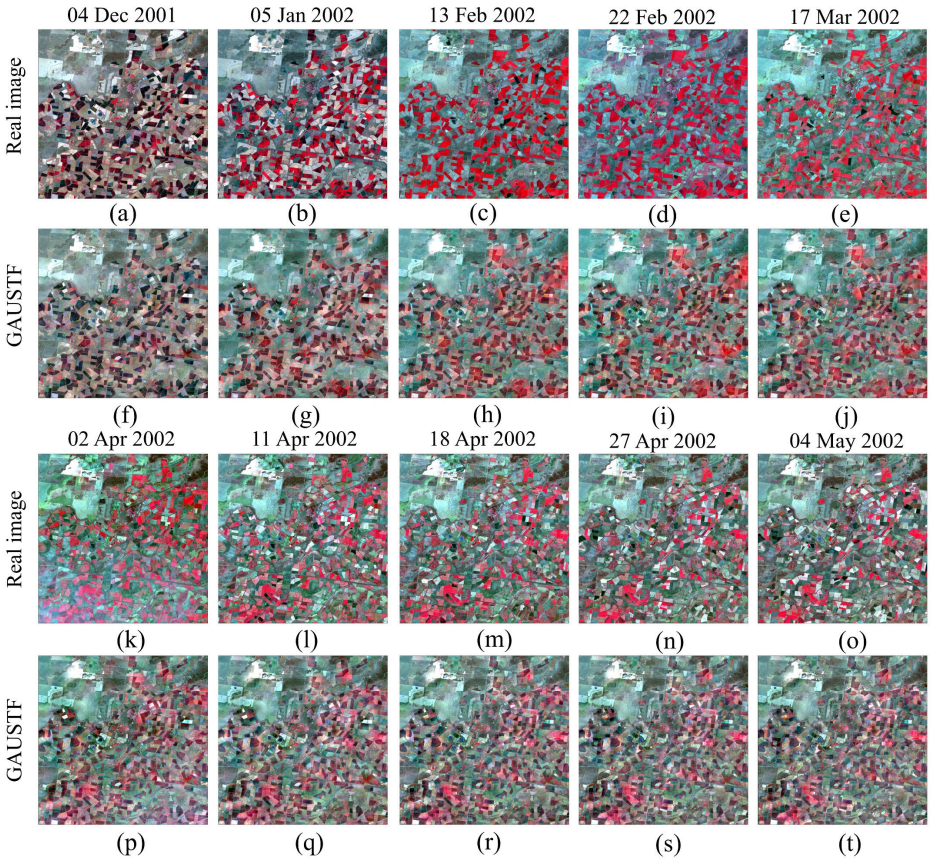


Fig. 7. Visual comparison of the time-series predicted results by GAUSTF against the actual Landsat images. (a)–(e) Actual Landsat images on December 4, 2001, January 5, 2002, February 13, 2002, February 22, 2002, and March 17, 2002. (f)–(j) GAUSTF predicted time-series for the same dates. (k)–(o) Actual Landsat images on April 2, 2002, April 11, 2002, April 18, 2002, April 27, 2002, and May 4, 2002. (p)–(t) GAUSTF predicted time-series for the same dates.

capture the phenological shift from bare land to growing crops; however, FSDAF yields a slightly higher reflectance in the NIR band, aligning more closely with the ground truth observation.

Overall, while all three methods produce visually comparable results at a macroscopic scale, their performance diverges significantly in complex local details. STARFM suffers from a noticeable block effect. FSDAF performs well in detecting phenological changes in vegetation, particularly in the NIR band, but tends to over-smooth structural details. In contrast, GAUSTF achieves a superior and more balanced performance in complex land-cover change scenarios. Relying on a mathematically sound Gaussian assumption within clustered homogeneous regions, GAUSTF uniquely excels at detecting the appearance and disappearance of distinct objects while strictly maintaining spatial boundaries. Quantitatively, STARFM achieves results comparable to FSDAF across most bands; however, similar to our proposed framework, its accuracy in the NIR band remains suboptimal. Notably, STARFM exhibits robustness in long-term predictions because, unlike FSDAF and GAUSTF, it does not rely on an initial clustering map. Long-term predictions inherently involve more frequent parcel-level land-cover transitions, making model performance highly sensitive to clustering map inaccuracies. For instance, empirical tests revealed that GAUSTF yields improved overall predictions for March 10, 2002, when utilizing a four-class clustering map rather than the three-class configuration currently employed. Nevertheless, optimizing GAUSTF's predictive accuracy in the NIR band remains an important direction for future research.

B. Assess Time-Series Prediction in the CIA Site

To evaluate GAUSTF's capability in reconstructing reflectance time series, we utilized the base image from November 25, 2001 to predict all subsequent dates within the dataset. This assessment primarily focused on spectral prediction fidelity. A visual comparison against the ground-truth observations is presented in Fig. 7.

The visual analysis of the GAUSTF-predicted time series demonstrates the method's effectiveness in capturing the macroscopic trends of crop phenological dynamics. Corresponding visual comparisons for STARFM and FSDAF are detailed in the Supplementary Material (Figs. SA3 and SA4). Furthermore, the quantitative evaluations of the three methods are illustrated in Fig. 8. As depicted, GAUSTF generally yields highly competitive results, positioning itself between STARFM and FSDAF. Notably, in the NIR band, GAUSTF and FSDAF achieve comparable accuracy, both significantly outperforming STARFM. While GAUSTF's long-term reflectance prediction capacity is secondary to that of FSDAF, it demonstrates a clear superiority over STARFM.

C. Single Pairs All-Around Performance in the LGC Site

The evaluation on the LGC dataset was specifically designed to assess the algorithms' robustness in predicting abrupt

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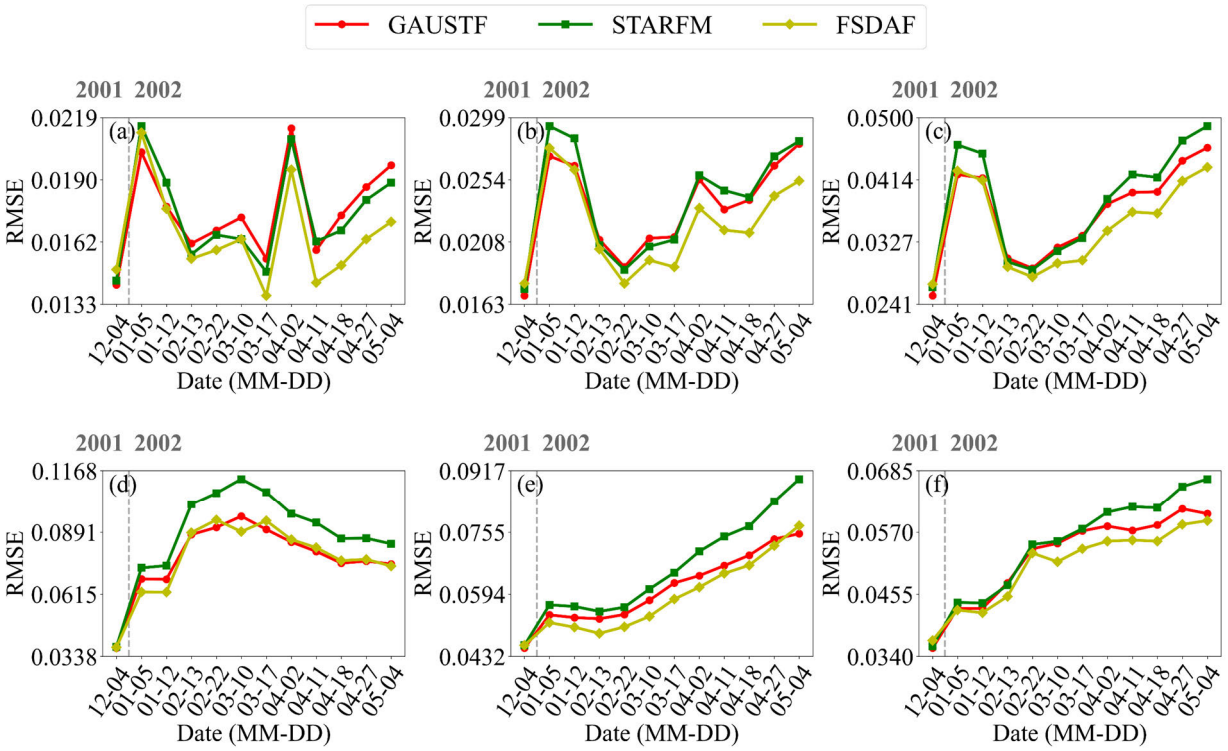


Fig. 8. Quantitative accuracy comparison of the long-term prediction capabilities of GAUSTF, STARFM, and FSDAF, measured by RMSE against the actual Landsat images of the corresponding dates. (a) Blue band. (b) Green band. (c) Red band. (d) NIR band. (e) SWIR 1 band. (f) SWIR 2 band.

land-cover changes. Subsequent to our base date (November 26, 2004), extensive inundation occurred across the region, triggering significant land-cover transformations. Using the base image from November 26, 2004, we predicted the surface reflectance for December 12, 2004. The base high-resolution image was classified into four spectral classes via K-means clustering. For these evaluations, the MODIS data were bilinearly resampled to a 500-m spatial resolution, which aligns with their native observational scale. These 500-m data were fed directly into GAUSTF, which operates optimally at this coarse resolution. In contrast, STARFM and FSDAF utilized MODIS data further resampled to 25 m. For FSDAF, the ISODATA clustering parameters were constrained to a minimum of 5 and a maximum of 6 classes, retaining default values for all other settings.

Table III presents the quantitative accuracy assessments for the three STF methods. While FSDAF achieved the lowest RMSE across most bands, it did not consistently dominate the AD and CC metrics. Notably, STARFM and GAUSTF demonstrated highly comparable overall spectral prediction performance at the LGC site. Their mean CC, AD, and RMSE values aligned closely, with GAUSTF's mean AD even slightly outperforming that of FSDAF.

Visually, as shown in Fig. 9, FSDAF produced noticeably smoother results compared to STARFM and GAUSTF. While STARFM continued to suffer from a pronounced block effect, FSDAF generated a more diffused prediction over the newly inundated areas, yielding brighter and more homogeneous intraobject colors. This smoothing behavior is likely attributable to its final window-based filtering step. Furthermore, as indicated in Table III, the LBP scores for FSDAF and GAUSTF were comparable and significantly superior to those of STARFM, highlighting their effectiveness in preserving object textures. Although STARFM yielded higher Edge detection scores, this metric was artificially inflated

TABLE III
ACCURACY ASSESSMENT OF THREE DATA FUSION METHODS APPLIED TO THE STUDY SITE LGC, PREDICTION OF DATE DECEMBER 12, 2004 USING THE DATE NOVEMBER 26, 2004. THE UNITS ARE REFLECTANCE

Band	CC	AD(10^4)	RMSE	Edge	LBP
			STARFM		
Blue	0.6677	-35.87	0.0171	**-0.1483**	0.2082
Green	0.6534	-18.47	0.0237	**-0.1194**	0.1927
Red	0.6614	-2.36	0.0297	**-0.1429**	**-0.0145**
NIR	0.7827	135.53	0.0413	-0.3933	0.0416
SWIR1	0.7329	**268.93**	0.0667	**-0.1840**	0.0909
SWIR2	0.6969	**297.21**	0.0568	**-0.1835**	0.0775
absmean	0.6992	**126.40**	0.0392	**0.1952**	0.1042
			FSDAF		
Blue	0.6630	**-29.52**	0.0178	-0.3641	**0.0214**
Green	**0.6617**	**-10.75**	**0.0226**	-0.4142	-0.0911
Red	**0.6844**	5.28	**0.0272**	-0.4178	0.0499
NIR	**0.8034**	144.10	**0.0396**	-0.4475	-0.0880
SWIR1	**0.7606**	281.61	**0.0636**	-0.5059	**-0.0260**
SWIR2	**0.7365**	307.62	**0.0541**	-0.4488	**-0.0706**
absmean	**0.7183**	129.81	**0.0375**	0.4331	0.0578
			GAUSTF		
Blue	**0.6775**	-36.12	**0.0164**	-0.3263	0.0962
Green	0.6596	-19.10	0.0231	-0.3497	**0.0398**
Red	0.6630	-2.31	0.0290	-0.3500	0.1579
NIR	0.7790	**134.74**	0.0411	**-0.3887**	**-0.0073**
SWIR1	0.7311	273.96	0.0679	-0.4467	0.0301
SWIR2	0.6963	300.89	0.0574	-0.3656	0.0108
absmean	0.7011	127.85	0.0392	0.3712	**0.0570**

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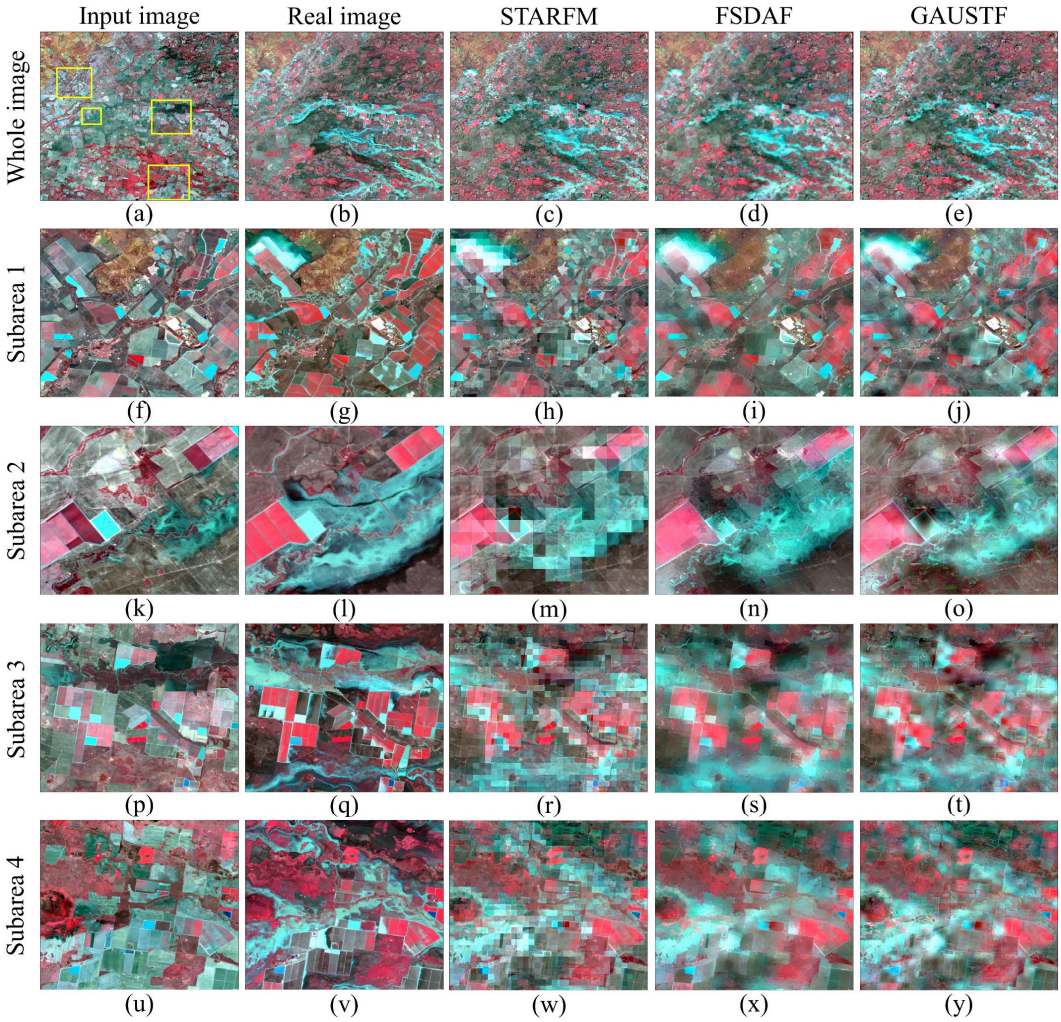


Fig. 9. Actual Landsat images of the LGC site acquired at t_1 and t_2 are shown in (a) and (b), respectively; the predictions for 12 Dec 2004 by STARFM, FSDAF, GAUSTF are shown in (c)–(e). (f)–(j) Show the input image, real image, and predictions by STARFM, FSDAF, and GAUSTF for Subarea 1, respectively.(k)–(y) Show the corresponding detailed images for Subarea 2, Subarea 3, and Subarea 4, respectively. Subareas 1 through 4 correspond to the upper, middle-left, middle-right, and lower yellow rectangles of the entire image, respectively.

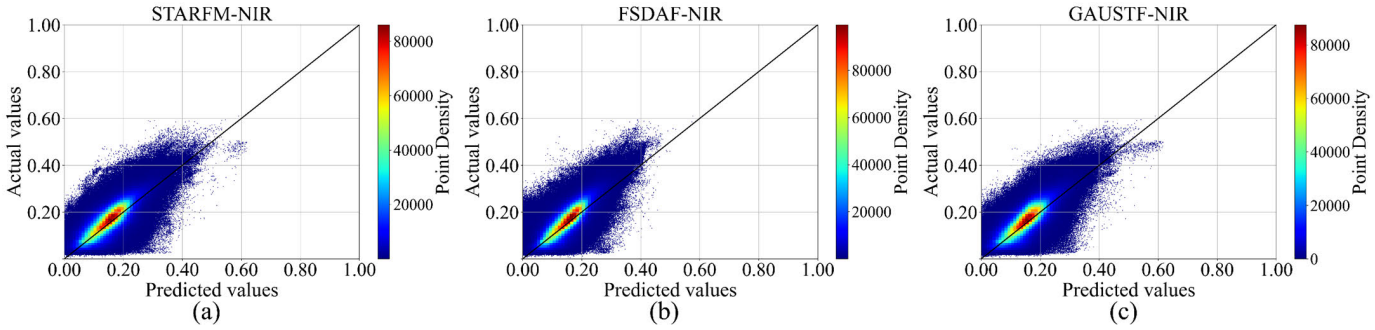


Fig. 10. Scatter plot of NIR band for the three methods in the LGC experiment: (a) STARFM, (b) FSDAF, and (c) GAUSTF.

by its inherent block artifacts rather than true structural fidelity.

The scatter plots for the NIR band predictions (Fig. 10) further corroborate these findings. FSDAF's predictions converge most tightly along the 1:1 reference line. GAUSTF exhibits a slightly better convergence than STARFM, though the margin is narrow. Scatter plots for the remaining bands are provided in the Supplementary Material (Fig. SA5).

These results demonstrate that by integrating change detection through uncertainty modeling, GAUSTF delivers versatile and highly reliable predictions even in scenarios of severe land-cover transformation. Its overall spectral fidelity is commensurate with STARFM, and both remain highly competitive with FSDAF, which is currently the most powerful spectral predictor among the three baselines.

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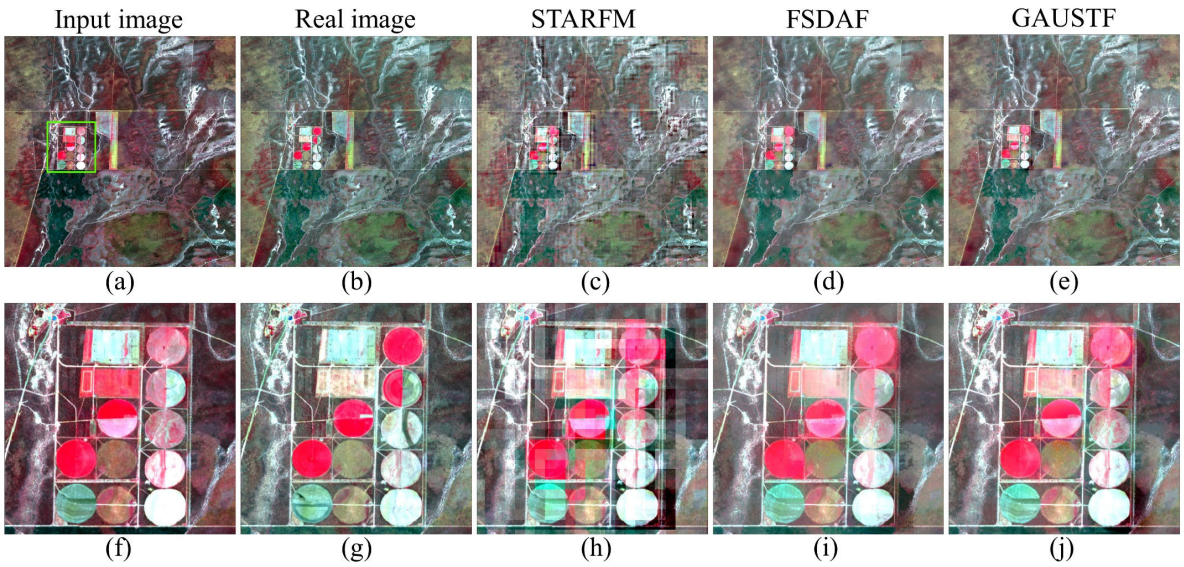


Fig. 11. Original Landsat images of the Waterbank site acquired at t_1 and t_2 are (a) and (b), respectively; the predictions for 18 Jun 2019 by STARFM, FSDAF, and GAUSTF are shown in (c)–(e). (f)–(j) show the input image, real image, and predictions by STARFM, FSDAF, and GAUSTF for the zoomed-in area of the green rectangle in the whole image, respectively.

TABLE IV
OVERLAY ANALYSIS STATISTICS OF DETECTED UNCERTAIN PIXELS VERSUS SVM-DERIVED CHANGED PIXELS

	User's accuracy (%)	Error rate (%)
Blue	23.7	60.7
Green	23.8	62
Red	25.9	49.3
NIR	16.5	69.4
SWIR1	17.8	49.1
SWIR2	15.8	46.1

User's Accuracy (Precision), defined as the proportion of flagged uncertain pixels that spatially coincided with the SVM-detected changed pixels

D. Effectiveness of the Uncertainty Evaluation in the LGC Site

The LGC dataset features significant land-cover transformations. To evaluate the effectiveness of our uncertainty module in capturing these dynamics, we analyzed the spatial distribution of the flagged "uncertain pixels." Across the six bands, an average of 40% of the total pixels were flagged as uncertain. The band-specific proportions were as follows: Blue (29.8%), Green (28.8%), Red (35.3%), NIR (33.4%), SWIR1 (51.6%), and SWIR2 (61.4%).

To quantitatively assess whether these statistical anomalies correspond to physical land-cover changes, we performed an overlay analysis. We first conducted a hard classification on the Landsat images from December 12, 2004 and November 26, 2004, using a support vector machine (SVM) to categorize the landscape into five types. Pixels that transitioned from nonwater to water were identified as "changed land-cover." We then calculated two metrics to evaluate the alignment between our statistical uncertainty and the actual land-cover changes: 1) user's accuracy (precision), defined as the proportion of flagged uncertain pixels that spatially coincided with the SVM-detected changed pixels; and 2) omission error rate, defined as the proportion of actual changed pixels that our module failed to flag as uncertain. The statistical results are summarized in Table IV.

GAUSTF achieves highly satisfactory reflectance prediction accuracy at the LGC site, particularly over the changed pixels. This success is fundamentally driven by our uncertainty modeling, which effectively captures the reflectance transitions. As shown in Table IV, the direct spatial overlap between uncertain pixels and the SVM-detected changes appears relatively moderate. However, this discrepancy is primarily due to the inherent misclassification errors and hard-boundary limitations of the SVM-based change detection used as a reference, rather than a failure of our uncertainty estimation.

E. Evaluation Using Sentinel-2 and Sentinel-3 Datasets

To evaluate the algorithm's performance in higher resolution scenarios, we applied GAUSTF to the fusion of Sentinel-2 and Sentinel-3 imagery, focusing on the Waterbank site. We selected May 14, 2019, as the base date to predict the reflectance on June 18, 2019, and September 11, 2019. For GAUSTF, we utilized a three-class clustering map. In FSDAF, the ISODATA clustering algorithm was configured with four classes; additionally, the number of pure pixels was set to 855, the block size to 20, and both the window size and the number of similar pixels to 20.

The Sentinel-3 data in the dataset are inherently upscaled to 10 m; we resampled these to 30 m using the majority (mode) method to preserve the original spatial values before feeding them into GAUSTF. For STARFM and FSDAF, we utilized the original Sentinel-3 data provided in the dataset.

A visual comparison for the first prediction date is presented in Fig. 11. All three methods successfully capture the overall macroscopic changes across the entire scene; however, detailed differences are evident. While FSDAF and GAUSTF yield visually comparable predictions, FSDAF more accurately

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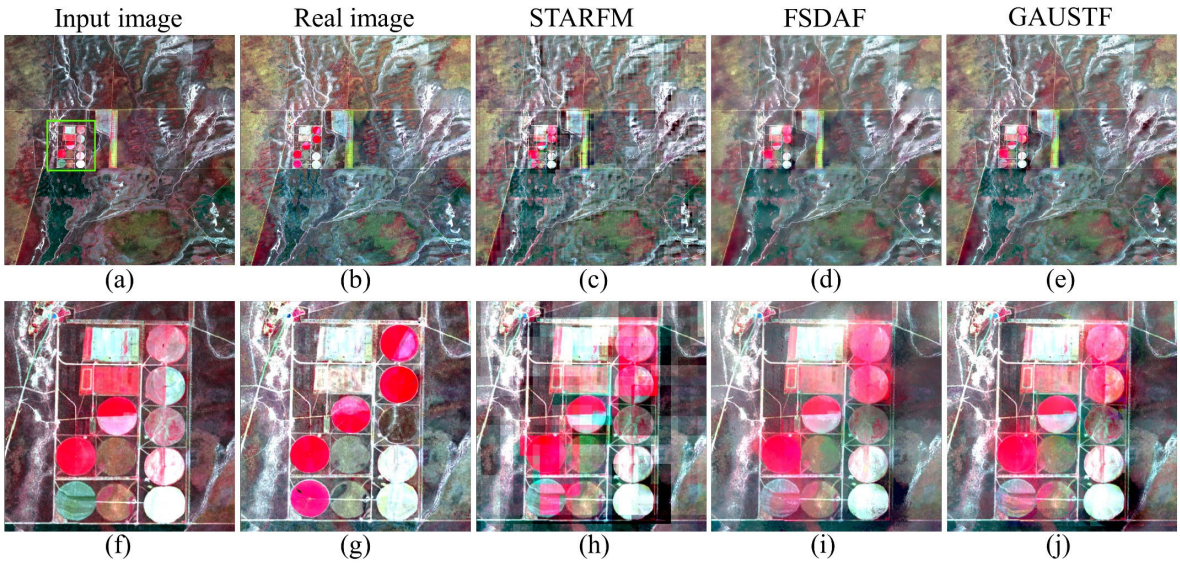


Fig. 12. Original Landsat images of the Waterbank site acquired at t_1 and t_2 are (a) and (b), respectively; the predictions for 11 Sep 2019 by STARFM, FSDAF, and GAUSTF are shown in (c)–(e). (f)–(j) show the input image, real image, and predictions by STARFM, FSDAF, and GAUSTF for the zoomed-in area of the green rectangle in the whole image, respectively.

captures fine-scale phenological changes, which are particularly noticeable within the central circled area. Conversely, STARFM suffers from a noticeable block effect, causing the predicted phenological changes within the circle to artificially spill over the boundaries—an artifact driven by the coarse footprint of the Sentinel-3 pixels. The quantitative comparisons are presented in Tables V and VI.

Prediction results for the second date are shown in Fig. 12. Although this target date is four months after the base date, the region did not experience significant land-cover or phenological changes. Visually, the three methods produce generally similar predictions, though STARFM yields a slightly sharper output compared to FSDAF and GAUSTF. In the zoomed-in region, the results generated by GAUSTF and FSDAF appear to be quite consistent with one another.

As shown in Table V, GAUSTF attains the highest CC values across all bands, and its RMSE values are essentially equal to those of FSDAF. In terms of AD, the three methods perform closely. STARFM achieves the highest scores in Edge detection, largely as a byproduct of its blocking artifacts, while GAUSTF performs best in the LBP metric, successfully retaining more structural texture details. The quantitative comparison for the zoomed-in region is provided in the lower section of Table V. While FSDAF yields the best RMSE for the first three bands, GAUSTF achieves the lowest RMSE for the NIR band. STARFM shows a marginally better AD in the Green and Red bands, but the overall AD values across the three methods remain highly comparable. Furthermore, the CC values of GAUSTF and FSDAF are similar and notably superior to those of STARFM. GAUSTF also registers the best LBP performance in this subarea.

As demonstrated in Table VI, GAUSTF achieves the highest CC values in the Blue, Green, and Red bands. FSDAF yields the smallest RMSE, though it remains essentially equivalent to that of GAUSTF. In terms of AD, the three methods again perform similarly. Overall, GAUSTF demonstrates spectral prediction capabilities comparable to FSDAF, and both definitively outperform STARFM. While STARFM performs best in Edge detection (inflated by block effects), GAUSTF consistently leads in the LBP metric for both the entire scene and the zoomed-in region.

F. Evaluation of Computational Efficiency

Computational efficiency is a critical metric for evaluating STF models, particularly when processing large-scale or long-term remote sensing datasets. Table VII presents the execution times (in seconds) of the five comparative algorithms across the CIA, LGC, and Waterbank study sites.

As indicated in the table, the proposed GAUSTF demonstrates an overwhelming advantage in computational efficiency, consistently ranking as the fastest algorithm across all scenarios. For instance, at the Waterbank site, GAUSTF required only 71 s to complete the fusion task, which is approximately 4.4 times faster than the second-fastest method (STARFM, 315 s) and orders of magnitude faster than FSDAF and Fit-FC. Conversely, RASTFM exhibited the highest computational burden, requiring up to 41 015 s (over 11 h) for the LGC site, despite the implementation of Numba JIT acceleration to optimize its spatial loops.

The consistent performance hierarchy (GAUSTF < STARFM < FSDAF < Fit-FC < RASTFM) across different spatial extents underscores the algorithmic lightweight nature of GAUSTF. By circumventing the exhaustive pixel-by-pixel moving window searches and complex optimization iterations typical of traditional methods, GAUSTF significantly reduces execution overhead, making it highly suitable for operational, large-area applications.

V. Discussion

A. Why GAUSTF Works Despite Non-Gaussian Reflectance

We sample pure pixels using a clustering map generated from the fine-resolution data at t_1. When utilizing Landsat-

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TABLE V

ACCURACY ASSESSMENT OF THREE DATA FUSION METHODS APPLIED TO THE STUDY SITE WATERBANK, PREDICTION OF DATE JUNE 18, 2019 USING THE DATE MAY 14, 2019. THE UNITS ARE REFLECTANCE

Entire image comparison														
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	0.8825	53.83	0.0081	**-0.0548**	0.0315	0.9318	**53.59**	**0.0068**	-0.1243	-0.0246	**0.9366**	53.79	0.0069	-0.0733	**0.0145**
Green	0.8900	49.37	0.0103	**-0.0267**	0.0282	0.9420	49.40	0.0077	-0.1194	-0.0964	**0.9486**	**49.34**	**0.0077**	-0.0588	**-0.0096**
Red	0.8730	57.96	0.0164	**-0.0363**	-0.0186	0.9279	58.35	0.0119	-0.1468	-0.1619	**0.9351**	**57.94**	**0.0118**	-0.0772	-0.0484
NIR	0.8885	**170.55**	0.0252	**-0.0349**	**0.0177**	0.9396	170.56	0.0209	-0.1254	-0.0842	**0.9445**	170.79	**0.0209**	-0.0985	-0.0222
absmean	0.8835	**82.93**	0.0150	**0.0382**	0.0240	0.9353	82.98	0.0118	0.1290	0.0918	**0.9412**	82.97	**0.0118**	0.0770	**0.0237**
Zoom-in comparison															
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	0.7876	54.28	0.0136	**-0.0144**	0.0500	**0.8382**	**53.97**	**0.0109**	-0.0869	-0.0301	0.8368	54.65	0.0116	-0.0220	**0.0233**
Green	0.8250	**45.64**	0.0192	**-0.0014**	0.0305	0.8790	46.73	**0.0143**	-0.0860	-0.0736	**0.8840**	45.93	0.0150	-0.0170	**-0.0017**
Red	0.8022	**47.45**	0.0337	**-0.0185**	**-0.0145**	**0.8629**	51.51	**0.0254**	-0.1185	-0.1045	0.8575	50.44	0.0275	-0.0325	-0.0340
NIR	0.8571	163.94	0.0450	**-0.0216**	0.0195	0.9124	**162.35**	0.0329	-0.1200	-0.0314	**0.9231**	165.49	**0.0327**	-0.0529	**0.0020**
absmean	0.8180	**77.83**	0.0279	**0.0140**	0.0286	0.8731	78.64	**0.0209**	0.1028	0.0599	**0.8754**	79.13	0.0217	0.0311	**0.0153**

TABLE VI

ACCURACY ASSESSMENT OF THREE DATA FUSION METHODS APPLIED TO THE STUDY SITE WATERBANK, PREDICTION OF DATE SEPTEMBER 11, 2019 USING THE DATE MAY 14, 2019. THE UNITS ARE REFLECTANCE

Entire image comparison														
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	0.8084	107.85	0.0136	**-0.1261**	0.0581	0.8451	**107.47**	**0.0128**	-0.1755	0.0119	**0.8515**	107.79	0.0129	-0.1280	**0.0462**
Green	0.8426	**118.83**	0.0167	**-0.1064**	0.0483	0.8866	119.06	**0.0149**	-0.1875	-0.0866	**0.8920**	118.93	0.0151	-0.1160	**0.0192**
Red	0.8232	**119.39**	0.0228	**-0.1250**	**-0.0328**	0.8695	119.99	**0.0191**	-0.2301	-0.1752	**0.8746**	119.76	0.0197	-0.1388	-0.0497
NIR	0.8183	21.10	0.0244	**-0.1333**	**0.0055**	**0.8680**	**21.06**	**0.0193**	-0.2247	-0.1199	0.8591	22.19	0.0203	-0.1692	-0.0086
absmean	0.8231	**91.79**	0.0194	**0.1227**	0.0362	0.8673	91.90	**0.0165**	0.2044	0.0984	**0.8693**	92.17	0.0170	0.1380	**0.0309**
Zoom-in comparison															
Band	STARFM					FSDAF					GAUSTF				
	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP	CC	AD ($\times 10^4$)	RMSE	Edge	LBP
Blue	0.6413	**111.57**	0.0197	**-0.0856**	0.0723	0.6788	111.99	**0.0176**	-0.1679	**-0.0139**	**0.6871**	112.01	0.0181	-0.1032	0.0460
Green	0.7076	**120.53**	0.0270	**-0.0669**	0.0486	0.7445	125.04	**0.0230**	-0.1811	-0.0637	**0.7568**	122.03	0.0240	-0.0901	**0.0160**
Red	0.6717	**124.93**	0.0456	**-0.0768**	0.0212	**0.7180**	131.55	**0.0381**	-0.2048	-0.0687	0.7109	128.00	0.0411	-0.1036	**0.0092**
NIR	0.7396	-46.24	0.0604	**-0.1672**	0.0246	0.7872	-42.91	**0.0500**	-0.2560	-0.0267	**0.7901**	**-37.97**	0.0506	-0.2068	**0.0155**
absmean	0.6901	100.82	0.0382	**0.0991**	0.0417	0.7321	102.87	**0.0322**	0.2024	0.0433	**0.7362**	**100.00**	0.0335	0.1259	**0.0217**

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TABLE VII

COMPARISON OF EXECUTION TIMES (IN SECONDS) FOR THE FIVE STF ALGORITHMS ACROSS DIFFERENT STUDY SITES

	GAUSTF	STARFM	FSDAF	Fit-FC	RASTFM
CIA	105	550	1660	3734	7189
LGC	219	1320	4170	9330	41015
Waterbank	71	315	1280	2312	8642

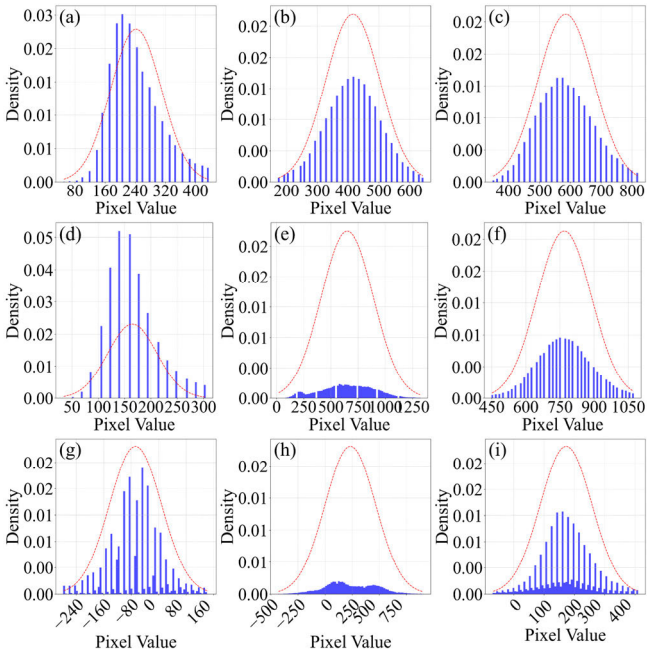


Fig. 13. Reflectance distributions for the three land-cover classes. (a)–(c) Sampled first-band reflectance from the fine-resolution data at the base date (November 25, 2001) and the corresponding fit Gaussian distributions for Classes 1, 2, and 3 based on the clustering map. (d)–(f) Sampled first-band reflectance from the fine-resolution data at the prediction date (January 12, 2002) and the corresponding Gaussian distributions for the three classes. (g)–(i) Distribution of the temporal reflectance difference (prediction date minus base date) and the corresponding Gaussian fits for the three classes.

MODIS pairs, for instance, the Landsat data alone cannot guarantee the existence of absolutely pure pixels within a coarse-pixel footprint. Consequently, this pure-pixel sampling serves merely as a coarse approximation of the true reflectance distribution for a given land-cover class. Furthermore, raw surface reflectance rarely follows a strict Gaussian distribution. Nevertheless, GAUSTF consistently achieves convincing estimations, primarily driven by the inherent statistical properties of temporal differencing. Specifically, the subtraction of two independent random variables yields a distribution that inherently converges toward a Gaussian form. In the derivation of GAUSTF, we utilize the difference between two approximated reflectance distributions. This operation generates a "residual distribution" that represents the temporal change in reflectance for a specific land-cover, mathematically exhibiting a much stricter adherence to a Gaussian distribution. Assuming the fine-resolution data accurately capture the true distribution of the land covers, a histogram comparison between the single-date fine-resolution distributions and the temporal difference distribution is presented in Fig. 13. This visual comparison clearly demonstrates how the subtraction process mathematically enforces a Gaussian approximation, thereby facilitating highly accurate reflectance predictions.

TABLE VIII

QUANTITATIVE DIVERGENCE METRICS MEASURING THE DEVIATION OF SAMPLED REFLECTANCE DISTRIBUTIONS FROM AN IDEAL GAUSSIAN FIT

	True distribution from Gaussian at t_1	True distribution from Gaussian at t_2	True distribution from Gaussian at t_2- t_1
		Class 1	
KL	1.5807	1.9610	0.9526
JS	0.4114	0.4744	0.2475
EMD	0.0124	0.0143	0.0094
BD	0.7740	0.9638	0.3261
		Class 2	
KL	1.3087	0.3654	0.0974
JS	0.3642	0.1172	0.0260
EMD	0.0110	0.0029	0.0013
BD	0.6536	0.1765	0.0268
		Class 3	
KL	1.3163	1.0524	0.4536
JS	0.3649	0.3069	0.1227
EMD	0.0110	0.0092	0.0055
BD	0.6554	0.5250	0.1416

Although most individual bands will not pass strict normality tests (such as the Kolmogorov–Smirnov test), we can quantitatively measure the degree to which the subtracted reflectance distribution approaches a Gaussian distribution using continuous divergence metrics, including Kullback–Leibler (KL) divergence, Jensen–Shannon (JS) divergence, Earth mover's distance (EMD), and Bhattacharyya distance (BD). Table VIII presents a comparison of these divergence metrics between the ideal Gaussian distribution and the sampled reflectance of the clustered classes at t_1, at t_2, and their temporal difference ($t_2 - t_1$).

Lower index values indicate that a distribution more closely approximates an ideal Gaussian form, as defined by its sample mean and standard deviation. As observed from both the histograms in Fig. 13 and the quantitative metrics in Table VIII, the subtracted reflectance data for each class exhibit a significantly closer approximation to a Gaussian distribution. This mathematical behavior fundamentally justifies why the derivation of GAUSTF is highly effective. Although the subtracted reflectance between t_2 and t_1 for a given class is not strictly Gaussian, it converges toward a Gaussian form much more closely than the original, single-date reflectance distributions at t_1 or t_2.

Second, according to the CLT, the linear combination of multiple independent, approximately Gaussian random variables converges toward a stricter Gaussian distribution. Consequently, a coarse mixed pixel is mathematically highly likely to follow a Gaussian distribution. In real-world applications, however, we cannot directly sample this distribution because each mixed pixel represents a single simulated Gaus-

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TABLE IX
PROPORTIONS OF PIXELS WITHIN THE 95% CONFIDENCE INTERVAL VERSUS DETECTED UNCERTAIN PIXELS FOR THE CIA DATASET

	Prediction for 12 Jan 2002		Prediction for March 10, 2002	
	within interval (%)	Uncertain pixels (%)	within interval (%)	Uncertain pixels (%)
Blue	79.88	20.12	87.25	12.75
Green	81.36	18.64	87.73	12.27
Red	81.51	18.49	86.47	13.53
NIR	80.86	19.14	85.62	14.38
SWIR1	73.87	26.13	77.91	22.09
SWIR2	75.09	24.91	78.82	21.18

TABLE X
PROPORTIONS OF PIXELS WITHIN THE 95% CONFIDENCE INTERVAL VERSUS DETECTED UNCERTAIN PIXELS FOR THE WATERBANK DATASET

	Prediction for June 18, 2019		Prediction for September 11, 2019	
	within interval (%)	Uncertain pixels (%)	within interval (%)	Uncertain pixels (%)
Blue	78.43	21.57	73.24	26.76
Green	80.51	19.49	74.94	25.06
Red	80.92	19.08	75.31	24.69
NIR	81.61	18.39	73.74	26.26

sian framework, and the observed satellite pixel value is merely one realization drawn from that distribution. Nevertheless, we can deduce this theoretical robustness directly from our original formulation. To illustrate, consider an extreme scenario where all coarse pixels contain only one homogeneous land-cover class. In this case, (6) simplifies to

$$y^{t_2}_{model} - y^{t_1}_{model} = x^{t_2} - x^{t_1}. \tag{20}$$

Consequently, (12) reduces to (proof omitted)

$$x^{t_2}_i - x^{t_1}_i = y^{t_2}_{obs} - y^{t_1}_{obs}. \tag{21}$$

This is evidently not an ideal estimation in practice, as it completely ignores spatial variance. Furthermore, this theoretical deduction reveals an important boundary condition: when the fine and coarse satellite images possess similar spatial resolutions, GAUSTF may yield suboptimal results compared to scenarios with a larger resolution gap. This limitation arises because the algorithm fundamentally relies on the spatial resolution differential to effectively capture and model the statistical RCs over time and space.

B. Proportion of Uncertain Mixed Pixels

The uncertainty analysis method introduced in Section II-D also provides a quantitative measure of reliability for our proposed model. If a coarse-resolution pixel can be accurately simulated using reflectance values drawn from a Gaussian distribution, then—in the absence of land-cover changes—the vast majority of these values should theoretically fall within the 95% confidence interval. Such adherence indicates that the construction and parameterization of the model (y_{model}) are statistically valid. To evaluate this, we analyzed the proportion of detected "uncertain pixels" across the entire image for both the CIA and Waterbank sites, as detailed in Tables IX and X.

As demonstrated in the tables above, the vast majority of coarse-pixel values in both the CIA and Waterbank datasets fall within the 95% confidence intervals established by the sampled Gaussian distributions. This confirms that, for the bulk of the image, the construction and parameterization of y_{model} are highly stable.

It is physically reasonable that the observed in-interval proportions are slightly lower than the theoretical 95% threshold. This discrepancy is primarily driven by the inherent complexities of real-world satellite observations, including sensor noise, residual atmospheric effects, and, most importantly, complex phenological variations occurring between the base and prediction dates. In this context, pixels falling outside the confidence interval (flagged as "uncertain pixels") represent specific areas where the simplified Gaussian assumption is challenged by rapid land-cover dynamics. Nevertheless, the fact that a large majority of pixels still adhere to the distribution confirms that the Gaussian approximation serves as an effective and appropriate mathematical foundation for the STF process.

It is also important to note that, compared to the LGC dataset (which experienced severe inundation), the CIA and Waterbank sites exhibit a significantly lower proportion of detected uncertain pixels. This observation indicates that although the flagged uncertain pixels may not perfectly overlap with hard-classified change boundaries, the detection mechanism effectively identifies and rectifies errors within the linear Gaussian combination model—specifically those induced by complex land-cover transformations and abrupt phenological shifts. Consequently, the statistical distributions presented here not only validate the internal consistency of the model but also robustly justify the necessity of the uncertainty rectification mechanism within the GAUSTF framework.

C. Sensitivity to the Number of Classes

We evaluated the algorithm's sensitivity to the predefined number of clustering classes by testing three-class and four-class configurations on the CIA dataset, as detailed in Table XI. Preliminary tests indicated that employing more than four classes resulted in an insufficient number of pixels being assigned to the additional classes, which fundamentally compromises the statistical robustness and effectiveness of the clustering process.

For the RMSE and CC metrics, the three-class configuration consistently outperforms the four-class configuration across all bands, with the most pronounced improvement observed in the NIR band. Although the four-class GAUSTF configuration yields slightly better AD values in specific bands (e.g., NIR and SWIR2), these differences are marginal. This suggests

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TABLE XI
GAUSTF PREDICTION COMPARISON USING THREE-CLASS AND FOUR-CLASS CLUSTERING MAPS FOR JANUARY 12, 2002 (BASE DATE: NOVEMBER 25, 2001) AT THE CIA SITE

	RMSE	AD ($\times 10^4$)	CC
		3 classes	
Blue	0.0177	27.89	0.7826
Green	0.0264	26.59	0.7884
Red	0.0416	27.87	0.8308
NIR	0.0681	-16.18	0.5064
SWIR1	0.0533	-63.36	0.8671
SWIR2	0.0429	-20.17	0.8784
		4 classes	
Blue	0.0180	27.96	0.7757
Green	0.0267	27.03	0.7827
Red	0.0422	28.23	0.8268
NIR	0.0726	-15.9	0.4672
SWIR1	0.0538	-60.76	0.8669
SWIR2	0.0425	-18.86	0.8788

TABLE XII
GAUSTF PREDICTION COMPARISON USING THREE-CLASS AND FOUR-CLASS CLUSTERING MAPS FOR MARCH 10, 2002 (BASE DATE: NOVEMBER 25, 2001) AT THE CIA SITE

	RMSE	AD ($\times 10^4$)	CC
		3 classes	
Blue	0.0173	76.17	0.7327
Green	0.0211	45.10	0.6834
Red	0.0320	22.44	0.7713
NIR	0.0962	22.93	0.2273
SWIR1	0.0578	-18.94	0.8308
SWIR2	0.0549	-11.86	0.8117
		4 classes	
Blue	0.0173	75.74	0.7144
Green	0.0217	43.27	0.6691
Red	0.0333	20.82	0.7534
NIR	0.1027	15.68	0.1476
SWIR1	0.0594	-19.04	0.8265
SWIR2	0.0561	-11.14	0.8033

that while increasing the number of classes may offer slight improvements in certain spectral aspects (such as AD), it does not significantly enhance overall accuracy and can degrade performance in structural and correlation metrics like RMSE and CC. The three-class configuration strikes a superior balance, effectively capturing landscape heterogeneity while ensuring reliable, stable statistical estimation for each class.

In the predictions for the second target date (Table XII), the four-class clustering input yields slightly better AD performance than the three-class input, with the exception of the SWIR1 band. However, the three-class configuration maintains superior RMSE and CC across the majority of the bands. This advantage is exceptionally prominent in the NIR band, where the three-class input significantly outperforms its four-class counterpart.

TABLE XIII
GAUSTF PREDICTION COMPARISON USING THREE-, FOUR-, AND SIX-CLASS CLUSTERING MAPS FOR DECEMBER 12, 2004 (BASE DATE: NOVEMBER 26, 2004) AT THE LGC SITE

	RMSE	AD ($\times 10^4$)	CC
		3 classes	
Blue	0.0164	-36.12	0.6725
Green	0.0231	-19.10	0.6492
Red	0.0290	-2.310	0.6540
NIR	0.0410	134.74	0.7828
SWIR1	0.0664	273.96	0.7173
SWIR2	0.0557	300.89	0.6770
		4 classes	
Blue	0.0164	-36.12	0.6775
Green	0.0231	-19.10	0.6596
Red	0.0290	-2.310	0.663
NIR	0.0411	134.74	0.7790
SWIR1	0.0679	273.96	0.7311
SWIR2	0.0574	300.89	0.6963
		6 classes	
	RMSE	AD ($\times 10^4$)	CC
Blue	0.0165	-35.9	0.6722
Green	0.0229	-18.61	0.6573
Red	0.0286	-2.01	0.6632
NIR	0.0420	135.25	0.7713
SWIR1	0.0651	274.34	0.7271
SWIR2	0.0542	300.29	0.6939

We further tested the algorithm's sensitivity to cluster quantities on the LGC dataset by comparing three-, four-, and six-class clustering maps. The results are summarized in Table XIII.

As observed in Table XIII, the performance differences introduced by varying the number of classes at the LGC site are virtually negligible.

To ensure generalizability, we extended this sensitivity evaluation to the higher resolution Sentinel-2 and Sentinel-3 datasets, testing three-, four-, and six-class configurations. The comparisons for the Waterbank site are presented in Tables XIV and XV.

For both prediction dates at the Waterbank site, the spectral prediction fidelity of GAUSTF remains remarkably consistent regardless of the number of classes utilized.

Synthesizing these comparisons, we observe that the algorithm's sensitivity to the input class quantity is more pronounced in bands exhibiting high temporal variance, such as the NIR band in the CIA dataset. Interestingly, while the NIR band at the LGC site also experiences significant variations, the majority of the land-cover change pixels were successfully flagged by the uncertainty module. This dynamic effectively mitigates the algorithm's reliance on the initial clustering map, rendering the predictions highly effective and

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TABLE XIV
GAUSTF PREDICTION COMPARISON USING THREE-, FOUR-, AND SIX-CLASS CLUSTERING MAPS FOR JUNE 18, 2019 (BASE DATE: MAY 14, 2019) AT THE WATERBANK SITE

	RMSE	AD ($\times 10^4$)	CC
		3 classes	
Blue	0.0069	53.79	0.9366
Green	0.0077	49.34	0.9486
Red	0.0118	57.94	0.9351
NIR	0.0209	170.79	0.9445
		4 classes	
Blue	0.0069	53.78	0.9368
Green	0.0077	49.29	0.9488
Red	0.0119	57.85	0.9352
NIR	0.0210	170.65	0.9461
		6 classes	
Blue	0.0069	53.77	0.9344
Green	0.0078	49.28	0.9459
Red	0.0123	57.93	0.9299
NIR	0.0209	170.39	0.9477

TABLE XV
GAUSTF PREDICTION COMPARISON USING THREE-, FOUR-, AND SIX-CLASS CLUSTERING MAPS FOR SEPTEMBER 11, 2019 (BASE DATE: MAY 14, 2019) AT THE WATERBANK SITE

	RMSE	AD ($\times 10^4$)	CC
		3 classes	
Blue	0.0129	107.79	0.8515
Green	0.0151	118.93	0.8920
Red	0.0197	119.76	0.8746
NIR	0.0203	22.19	0.8591
		4 classes	
Blue	0.0129	107.8	0.8523
Green	0.0151	118.85	0.892
Red	0.0197	119.61	0.8743
NIR	0.0205	21.67	0.858
		6 classes	
Blue	0.0129	107.83	0.8456
Green	0.0153	118.85	0.8826
Red	0.0203	119.59	0.8637
NIR	0.0207	20.88	0.8565

insensitive to the predefined number of classes in such abrupt-change scenarios.

Furthermore, it is worth noting that varying the number of clusters (e.g., from 3 to 6) has a negligible impact on the overall computational execution time of the algorithm. Therefore, determining the optimal number of classes is primarily dictated by the physical tradeoff between securing sufficient sample sizes for accurate statistical estimation and maintaining effective landscape class representation, rather than computational overhead constraints.

D. Incorporated Into Deep Learning Frameworks as a Prior Knowledge Constraint

Current state-of-the-art STF methods are increasingly moving toward a "physics-informed" paradigm. Purely data-driven deep learning models, while powerful, often suffer from poor generalization when applied to unseen landscapes or sensors due to their "black-box" nature. To address this, recent studies have begun to embed physical laws directly into network architectures.

An example is SwinSTFM [54]. This method maps the physical process of linear spectral mixing into the mathematical operations of the attention mechanism

$$\text{Attention}(Q,K,V) = Norm\left(\begin{matrix} SoftMax\left(\frac{Q_0 K_0^T}{\sqrt{d}} + B\right) \\ \cdot SoftMax\left(\frac{Q_p K_p^T}{\sqrt{d}} + B\right) \end{matrix}\right) V \tag{22}$$

where Q represents the learned coarse feature representations at the reference date, acting as the "mixed pixels" to be decomposed; K represents the learned fine feature representations at the reference date, acting as the "endmembers" to calculate the similarity with the coarse pixel; V represents the coarse features containing temporal info, serving as the total variation energy to be distributed to fine pixels; B represents a learnable correction term derived from the coarse images at both dates (t_1 and t_2), used to dynamically adjust the unmixing weights to account for land-cover changes and nonlinear variations.

Specifically, the essence of linear spectral mixing is structurally enforced in (22) through two key operations: First, the Norm(·) function imposes the physical sum-to-one constraint on the attention scores, effectively treating them as fractional abundances. Second, the final matrix multiplication with V realizes linear superposition, ensuring that the fused features are generated via a strictly weighted aggregation of temporal information, mirroring the physical formulation $C = \sum A_i F_i$.

The success of SwinSTFM in mapping physical equations to network operators sets a precedent, demonstrating that deep learning models gain superior generalization and interpretability when constrained by rigorous domain knowledge. Extending this paradigm from deterministic spectral mixing to probabilistic statistical inference, we envision a substantial potential in further hybridizing our GAUSTF model with deep representation learning.

Specifically, we envision substantial potential in further reformulating the fusion task not merely as image regression, but as a parameter estimation problem. In such a framework, a deep neural network would function as a dual-branch estimator to predict the sufficient statistics of the unobserved fine-resolution image: the prior mean ($\Delta\mu$) and the pixel-wise aleatoric uncertainty (σ^2). These predicted parameters would then serve as inputs to a nonlearnable analytical fusion layer. This layer executes the GAUSTF update rule as a differentiable hard constraint to generate the final prediction $\hat{F}$

$$\widehat{F} = \Delta\mu + K\left(\sigma^2\right) \cdot Up\left(C_{obs} - Down\left(\Delta\mu\right)\right). \tag{23}$$

Since the analytical operations involved—including convolution, upsampling, and element-wise arithmetic—are fully differentiable, this design preserves the gradient flow, enabling

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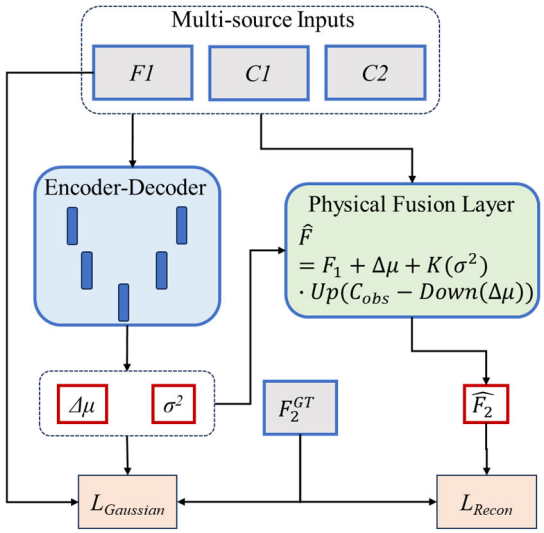


Fig. 14. Illustration of the proposed network based on GAUSTF.

the entire framework to be optimized end-to-end via standard backpropagation.

To train this physics-embedded architecture, a compound objective function is required to simultaneously ensure perceptual fidelity and statistical consistency. The total loss $\mathcal{L}_{total}$ would be formulated as

$$\mathcal{L}_{total} = \mathcal{L}_{Recon} + \lambda \cdot \mathcal{L}_{Gaussian}. \tag{24}$$

The first term, reconstruction loss, supervises the final fused output against the ground truth F_{GT} to enforce structural accuracy

$$\mathcal{L}_{Recon} = \left\| F_{pred} - F_{GroundTruth} \right\|_1. \tag{25}$$

The second term, Gaussian penalty (formulated as heteroscedastic negative log-likelihood), explicitly constrains the intermediate parameter estimation

$$\mathcal{L}_{Gaussian} = \frac{\|\Delta\mu - (F_{GroundTruth} - F_1)\|^2}{2\sigma^2} + \frac{1}{2}\log\sigma^2. \tag{26}$$

While $\mathcal{L}_{Recon}$ ensures the final prediction accuracy, $\mathcal{L}_{Gaussian}$ forces the network to learn a meaningful uncertainty distribution. This enables adaptive physical fusion. The model autonomously learns to down-weight the prior μ in regions with complex spectral changes (high σ^2) and rely more heavily on the observed residuals, combining the pattern-recognition power of deep learning with the rigorous physical interpretability of the GAUSTF model. This physics-informed paradigm enhances generalization on out-of-distribution data and prevents generative "hallucinations" by strictly bounding the output with observation residuals. The network architecture is illustrated in Fig. 14.

E. Other Related Works

There are several other works that improve the STF by implementing Gaussian distribution models. A representative study is the STRUM [56]. Similar to unmixing-based data fusion (UBDF) [61], it restrains the linear unmixing method using specific measurements. STRUM addresses this constraint by assuming a Gaussian distribution for each endmember; consequently, coarse reflectance is also a Gaussian distribution. Then, Bayesian theory is applied to constrain the unmixing estimation. Thus, the method achieves a statistical compromise between the information from the coarse observation and the unmixed information. STRUM is a well-designed method that demonstrates the powerful application of statistics in STF. However, hybrid methods often involve empirical steps, lack flexibility, and are difficult to expand.

In recent years, considering the limitation of STF method on the global application, two other potential methods are developed for global EO data generation. First, OL-based STF improvement methods [62] accelerate STF methods such as STARFM, Fit-FC, and FSDAF by refining pixel-wise neighbor similarity searches in the final stages. However, they cannot process object-level information through more complex operations, such as through thin plate spline (TPS) interpolation in FSDAF. In addition, their results are highly sensitive to segmentation quality, which can be time-consuming to optimize, especially if using deep learning methods like the segment anything model (SAM). Second, the ROBOT method [63], which is approximately 600 times faster than FSDAF, requires at least two fine-resolution datasets from adjacent dates to achieve accuracy comparable to Fit-FC. This makes it impractical for regions with frequent cloud cover or rain, where obtaining consistent fine-resolution data is challenging. We anticipate that future advancements in these methods, including GAUSTF, will address their current shortcomings and enable their effective application to the production of global high-spatiotemporal-resolution remote sensing imagery.

The theoretical foundations discussed above—specifically the residual distribution approximation and the CLT—indicate that GAUSTF possesses significant potential for broader scenarios beyond classic benchmark datasets. Primarily, the model is highly adaptable to extremely heterogeneous and fragmented landscapes, such as smallholder agricultural parcels in complex mountainous regions or dynamic urban-rural fringes. In these complex scenarios, traditional linear fusion methods often encounter bottlenecks due to unpredictable spatial mixing. In contrast, according to the CLT, higher subpixel fragmentation means a greater number of independent land-cover variables are linearly combined within a single coarse pixel. This extreme mixing paradoxically strengthens our Gaussian assumption, allowing the uncertainty mechanism to rectify subpixel variations more effectively. Furthermore, because the model benefits from a larger resolution difference between fine and coarse data to satisfy the CLT, GAUSTF is broadly applicable to fusing emerging multisensor datasets with massive spatial resolution gaps. Future applications could confidently extend this framework to complex ecological monitoring tasks, such as tracking vegetation degradation or land abandonment in highly heterogeneous topographies, where capturing subpixel dynamic changes is critical.

VI. Conclusion

In this study, we addressed the limitations of existing procedural STF methods by proposing GAUSTF, a novel framework that innovatively unifies deterministic linear spectral unmixing with stochastic Gaussian distribution assumptions. By transforming the complex fusion task into a rigorous parameter inference problem, GAUSTF overcomes the rigidity of

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traditional heuristic-based algorithms. Comprehensive evaluations across diverse benchmarks—including heterogeneous agricultural lands, dynamic flood scenarios, and multisensor Sentinel-2/Sentinel-3 pairs—validate the efficacy of this unified statistical model. Specifically, GAUSTF achieves spectral prediction accuracy comparable to the widely adopted FSDAF benchmark and consistently outperforms STARFM, while significantly mitigating "block effects" and "over-smoothing" issues through its object-oriented texture preservation mechanism. Most notably, the model demonstrates exceptional computational efficiency, operating approximately 16–19 times faster than FSDAF. Ultimately, the primary contribution of this work lies in establishing a statistically explicit, mathematically transparent, and computationally lightweight framework. Its closed-form structure not only offers a scalable solution for global monitoring but also provides a robust, differentiable physical constraint layer, paving the way for the integration of statistical physics into the next generation of deep learning architectures in remote sensing.

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